Controlling Shareholders and Supermajority Rules*

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Abstract

I examine how voting rules shape information transmission between controlling shareholders and minorities in conflicted transactions. I develop a model of costly signaling where misrepresentations that harm minorities carry litigation risk, while conservative reporting is relatively cheap—reflecting the realities of corporate litigation. This asymmetric cost structure generates a trade-off: controlling shareholders must balance the private benefits from value-destroying projects against the costs of misrepresentation needed to secure minority approval. I show that the majority of the minority (MoM) rule exhibits superior informational properties compared to both lenient and moderately stringent supermajority rules. The key insight is that MoM occupies a "knife-edge" position: relative to the MoM, lenient rules encourage controllers with bad projects to lie because approval is cheap, while stringent rules induce controllers with good projects to withhold information because approval is too costly. MoM minimizes the pooling equilibrium region where truthful and deceptive communications cannot be distinguished. However, sufficiently stringent supermajorities can achieve complete separation and truthful disclosure, at the cost of blocking valuable transactions. These findings provide a novel informational rationale for the widespread use of MoM in corporate law.

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1 Introduction

The rise of controlling shareholders represents one of the most significant developments in corporate governance over the past two decades. Even in traditionally diffuse-ownership jurisdictions like the United States and United Kingdom, controlling shareholders have expanded their influence, particularly in very large firms that were once bastions of dispersed ownership (Aminadav and Papaioannou, 2020; Holderness, 2009; Pargendler, 2019; Tallarita, 2024).[1] This shift has sparked intense debate among legal scholars and practitioners, especially following Delaware's 2025 SB 21 reforms, which fundamentally altered the landscape of controlling shareholder transactions. By eliminating the dual requirements for approval by both an independent special committee and a majority of minority shareholders established in *Kahn v. M&F Worldwide Corp.*, 88 A.3d 635 (Del. 2014), and providing de facto immunity to blockholders below 33% of voting power, these reforms have reignited fundamental questions about minority shareholder protection (Bainbridge, 2025; Whitehead, 2025; Khoo and Tallarita, 2025). The implications extend beyond the U.S., as other jurisdictions grapple with similar tensions between facilitating efficient transactions and protecting minority shareholders (Ventoruzzo, 2015; Enriques, 2014; Lim, 2018).

The dominance of controlling shareholder structures is not merely a U.S. phenomenon but reflects global corporate reality. Recent OECD data reveals that concentrated ownership remains the prevailing governance structure across most economies, with controlling shareholders present in the majority of listed companies worldwide (OECD, 2023; De La Cruz et al., 2019). This pattern holds across diverse legal and economic systems: La Porta et al. (1999) document the prevalence of controlling shareholders across 27 wealthy economies; Claessens et al. (2000) find concentrated ownership in over two-thirds of East Asian corporations; while Faccio and Lang (2002) demonstrate that family control dominates Western European firms outside the UK and Ireland.

Despite the economic significance of the agency costs between controlling and minority shareholders—what scholars have termed "Principal-Principal" conflicts—corporate law theory remains disproportionately focused on the board-shareholder agency problem (Goshen and Squire, 2017; Kraakman and Armour, 2017; Anabtawi, 2005). The canonical works in corporate governance—from Bebchuk (2005)'s shareholder empowerment thesis, Blair and Stout (1999)'s team production model, and Bainbridge (2003)'s director primacy theory—primarily address conflicts between professional managers and dispersed shareholders. This theoretical emphasis persists despite empirical evidence that controlling shareholder agency costs may dwarf traditional managerial agency costs in most of the world's corporations (Johnson et al., 2000; Djankov et al., 2008; Atanasov et al., 2014).

When scholars do examine controlling shareholder agency costs, the theoretical framework has largely centered on mechanisms to prevent extraction of private benefits of control (Gilson, 2006; Gilson and Gordon, 2003; Goshen, 2003; Gilson and Schwartz, 2015). The empirical literature documents substantial private benefit extraction: Dyck and Zingales (2004) demonstrate that control premiums average 14% globally but vary dramatically across countries, from -4% in Japan

[1] For instance, Holderness (2009) shows that concentrated ownership is more common than typically assumed in the United States, particularly among smaller public companies.

to 65% in Brazil; Barclay and Holderness (1989) find significant block premiums in U.S. negotiated trades; while Albuquerque and Schroth (2008) show how legal institutions affect the magnitude of private benefits. Yet the effectiveness of legal constraints remains contested. As Choi (2018) argues, controlling shareholders' long-term horizons and reputational concerns may naturally align their interests with minority shareholders (see also Anderson and Reeb, 2003; Villalonga and Amit, 2006). Moreover, controlling shareholders can provide valuable monitoring that reduces managerial agency costs (Shleifer and Vishny, 1986; Burkart et al., 2003), making the necessity and design of legal interventions fundamentally an empirical question.

This paper provides a novel angle to this debate by examining the informational effects of voting mechanisms used to govern conflicted *controller* transactions. Beyond ex-post litigation—whose effectiveness remains debated (Thompson and Thomas, 2016; Choi and Pritchard, 2009)—corporate law relies heavily on ex-ante voting requirements for fundamental corporate changes: mergers and acquisitions (Restrepo, 2021), substantial asset sales , capital structure alterations (Gompers et al., 2010), and most crucially, conflicted transactions involving controlling shareholders (Kraakman and Armour, 2017). The centrality of conflicted transactions is evident in recent Delaware jurisprudence, for instance, in *In re Match Group, Inc. Derivative Litigation*, 2024 WL 1146412 (Del. Ch. Mar. 13, 2024) and *Tornetta v. Musk*, 310 A.3d 430 (Del. Ch. 2024). These transactions represent a first-order concern precisely because they create the starkest conflicts between controlling and minority shareholders (Restrepo, 2021; Djankov et al., 2008).

The standard mechanism for "cleansing" conflicted transactions requires approval by non-conflicted shareholders, effectively excluding the controlling shareholder from voting (Subramanian, 2005). This "majority of the minority" (MoM) rule has become a cornerstone of minority protection across jurisdictions (Goshen and Hamdani, 2019). However, corporate law employs various other voting thresholds that functionally resemble MoM but operate through different mechanics (Easterbrook and Fischel, 1983). Special resolutions in the United Kingdom (UK), requiring 75% approval, create varying effective thresholds depending on the controller's stake—if a controlling shareholder holds 60% of shares (assuming one-share-one-vote), minorities need only garner 15% support, making the requirement less stringent than MoM.[2] Conversely, supermajority rules requiring extreme consensus, such as the 90% threshold for compulsory share acquisitions in squeeze-out transactions for the UK and Hong Kong, can demand greater minority support than simple MoM.[3] These variations in voting rules may have profound implications for corporate governance outcomes.

While prior theoretical research has focused on the trade-offs between MoM and supermajority rules through the lens of Type I and Type II errors—approving bad transactions versus blocking good ones (Thompson, 1995)—this literature predominantly focuses on settings where information asymmetry plays a secondary role (Maug and Rydqvist, 2009). Other theoretical literature on voting has explored various mechanisms, from Harris and Raviv (1988) on one-share-one-vote to Grossman and Hart (1988) on security voting structures, but has largely abstracted from the

[2]See Section 283 of the UK Companies Act. Australia similarly requires "Special Resolutions" with 75% of eligible votes for actions such as modifying the company Constitution or selective share reductions (See Section 9 read with Section 250MA of the Australian Companies Act).

[3]See Section 979 of the UK companies Act and Section 694 of the Hong Kong Companies Ordinance.

strategic communication between informed controllers and uninformed minorities. My paper departs from this approach by placing information transmission at the center of the analysis. In my framework, voting rules fundamentally shape how controlling shareholders choose to communicate private information to uninformed minority shareholders, building on insights from the cheap talk literature (Crawford and Sobel, 1982; Farrell and Rabin, 1996) and costly signaling models (Spence, 1973; Milgrom and Roberts, 1986).

I present a model of adverse selection with costly signaling, where the cost structure exhibits a crucial asymmetry that reflects litigation realities. Misrepresentations that harm minority shareholders—overstating project quality or understating conflicts—carry substantial litigation risk and reputational costs (Dyck et al., 2010). Delaware courts have increasingly scrutinized disclosure quality in controller transactions, as seen in *In re Tesla Motors, Inc. Stockholder Litigation*, 298 A. 3d 667 (Del: Supreme Court 2023). Conversely, conservative reporting or non-disclosure typically generates no actionable losses for minorities, making it relatively cheap—a pattern consistent with the business judgment rule's protection of good faith decisions (Bainbridge, 2004). This asymmetry creates complex strategic considerations: controlling shareholders must balance the benefits of pushing through value-decreasing projects that yield private benefits against the costs of misrepresentation needed to secure approval. In equilibrium, some controllers with genuinely value-creating projects choose conservative reporting to minimize litigation risk, while others with value-destroying projects may find it optimal to misrepresent when private benefits are substantial. Critically, such lying induces pooling equilibria where minority shareholders cannot distinguish truthful from deceptive communications, fundamentally undermining the information transmission function of corporate disclosures (Dye, 1985; Verrecchia, 1983; Choi and Spier, 2024).

The model reveals that the majority of the minority rule possesses remarkable informational efficiency properties. Relative to both lenient supermajorities and stringent supermajorities (up to a critical threshold), MoM minimizes the region of pooling equilibria, maximizing truthful disclosure. The intuition reflects MoM's "knife-edge" position: lenient supermajorities encourage controllers with bad projects to lie because securing approval is cheap, while stringent supermajorities induce controllers with good projects to withhold information because the communication required for approval is prohibitively costly. These results complement related work by Levit et al. (2023), who focus on information aggregation among dispersed voters, and Levit et al. (2024), who examine how trading opportunities affect voting outcomes. MoM optimally balances these forces, though notably, sufficiently stringent supermajorities can achieve complete separation where all types communicate truthfully—at the cost of blocking many value-creating transactions, consistent with Gilson and Gordon (2003)'s analysis of inefficient minority protection.

This paper contributes to several strands of literature. Unlike models of strategic voting and information aggregation (Levit et al., 2023, 2024; Meirowitz and Pi, 2022), which focus on coordination problems among dispersed voters or the relationship between trading and voting, my framework emphasizes the strategic communication between an informed controlling shareholder and uninformed minorities. While papers like Meirowitz and Pi (2023) analyze voting with endogenous information acquisition, I focus on how voting rules shape disclosure incentives. In my paper,

the controlling shareholder's *communication strategy* drives equilibrium outcomes and efficiency properties. My work also relates to the literature on optimal voting mechanisms (Gershkov et al., 2017) and the design of corporate voting rules (Easterbrook and Fischel, 1983; Miller, 2021), but emphasizes the informational rather than purely distributive consequences of different thresholds (Harris and Raviv, 2010).

My paper also builds on prior work on MoM and supermajority rules, which reports varied findings on MoM's effectiveness. In the United States, case studies of Majority-of-the-Minority (MoM) approvals in buyouts conclude that these rules "did not hurt, but did not do much good either." In particular, active shareholders who opposed deals on the ground that the price was too low were often unable to block or renegotiate them (Rock, 2018). Critics also point to empirical evidence from countries like Israel, which yields mixed results regarding MoM's impact. For example, one study finds that minority voting power did not significantly influence institutional investors' decisions, nor did they consistently object to suspected value-diversion proposals (Hamdani and Yafeh, 2013). Moreover, "frictions" such as conflicts-of-interest, information asymmetry, and "empty voting" can impede the quality of shareholder voting under MoM, prompting proposals for alternative mechanisms like "Majority of the Quality Shareholders" (Cunningham, 2021; Hu and Hamermesh, 2025).

Despite these concerns, the MoM rule is also described as a "promising tool" in contexts such as freeze-out mergers, where it can facilitate a fair price negotiated by an independent Special Negotiating Committee and acceptable to minority shareholders. On this view, it is more efficient and superior to coercive tender offers followed by short-form mergers, which can leave minority shareholders with less valuable stub shares (Booth, 2013). In the context of China, empirical work suggests that MoM disapprovals for related-party transactions (RPTs) in listed companies are strongly associated with efficiency gains—particularly for over-investing firms—even when proposals are not outright rejected (Cao et al., 2024).[4]

The remainder of the paper proceeds as follows. Section 2 presents the model setup, detailing the information structure and cost assumptions. Section 3 characterizes the equilibrium under different voting rules. Sections 4 and 5 analyze informational efficiency and welfare properties of various thresholds, while Section 6 concludes.

2 The Model: Primitives and Assumptions

In this section, I characterize the key model that motivates the rest of my analysis. Subsequent theorems or proof will cite only the numbered statements below.

[4]However, other work suggests that MoM might be less effective for RPT governance in China due to "investment myopia" among minority shareholders, who can be influenced by short-term stock price performance and pre-vote earnings management (Lin et al., 2025).

2.1 Controlling Shareholders

2.1.1 Primitives

There exists a single *controlling shareholder* (CS) who permanently owns a strictly positive but non-unanimous voting stake $\alpha \in (0,1)$, while a large but finite number N of symmetric minority shareholders–which will be discussed later–hold the residual $1-\alpha$.

The quality of a project is denoted by the variable θ. It is realised once, observed privately by the CS, and then remains fixed throughout the game. Importantly, θ is distributed $\theta \sim F$, and F is continuous on the compact interval $\left[\underline{\theta}_0, \overline{\theta}_0\right] \subset \mathbb{R}$, where the support endpoints satisfy

$$[\,\underline{\theta}_0,\ \overline{\theta}_0], \qquad \underline{\theta}_0 < 0 < \overline{\theta}_0. \tag{P1}$$

The density $f := F'$ exists, is strictly positive on the support, and is *log–concave*:

$$f(x) > 0 \quad \text{and} \quad \log f(\cdot) \text{ is concave on } [\underline{\theta}_0, \overline{\theta}_0]. \tag{P2}$$

Let $\mathcal{M}(f) := \arg\max_{x\in[\underline{\theta}_0,\overline{\theta}_0]} f(x)$ be the (possibly non-singleton) set of modes. We assume $0 \in \mathcal{M}(f)$. By log-concavity, f is unimodal (single-peaked); hence f is weakly nondecreasing on $[\underline{\theta}_0, 0]$ and weakly nonincreasing on $[0, \overline{\theta}_0]$. Log-concavity supports single-crossing/monotone posterior arguments (Bagnoli and Bergstrom (2005)); the zero-mode refinement is used only in pooling–mass comparisons across rules.

Implementing the project gives the controller transferable private benefit $B(\theta)$ on top of the pro-rata cash flow $\alpha\theta$. We assume

$$B \ :\ [\,\underline{\theta}_0, \overline{\theta}_0] \longrightarrow \mathbb{R}, \qquad B \in C^1, \qquad 0 < B'(\theta) < 1-\alpha \quad \big(\forall\, \theta\big). \tag{P3}$$

where the level can be normalized by setting $B(\underline{\theta}_0) = 0$.

Because the marginal diversion is always strictly below the minorities' residual cash-flow share due to assumption P3, the net minority cash flow

$$(1-\alpha)\,\theta \ - \ B(\theta)$$

is strictly increasing over the entire support $[\underline{\theta}_0, \overline{\theta}_0]$, a property used in all single-crossing and equilibrium-selection arguments that follow.

2.1.2 Signalling technology

After observing θ the CS sends a publicly observable real message $m \in \mathbb{R}$. Sending the message incurs an out-of-pocket cost $C(m, \theta)$. I impose four axioms which together formalise an *asymmetric (one-sided) convex signalling cost.*

Continuity and differentiability: C is continuous on $\mathbb{R}^2$. It is C^1 off the diagonal and twice continuously differentiable on each open wedge:

$$\begin{gathered} C \in C^1\big(\{(m,\theta) \in \mathbb{R}^2 \,:\, m \neq \theta\}\big), \\ C \in C^2\big(\{(m,\theta) \in \mathbb{R}^2 \,:\, m > \theta\}\big) \text{ and } C \in C^2\big(\{(m,\theta) \in \mathbb{R}^2 \,:\, m < \theta\}\big). \end{gathered} \tag{S1}$$

Here, "off the diagonal" refers to the region outside the diagonal $\{(m,\theta) \in \mathbb{R}^2 \,:\, m = \theta\}$ (i.e. at any point with $m \neq \theta$). By C^1 (resp. C^2) on a region, the definition is that all first (resp. first and second) partial derivatives exist and are continuous on that region.

Cost of reporting true valuation: Reporting the exact quality never consumes resources:

$$C(\theta,\theta) = 0. \tag{S2}$$

Asymmetric marginal-cost profile: For every true quality θ define $\Delta := m - \theta$. The first derivative of the cost satisfies

$$C_m(m,\theta) \;=\; \begin{cases} -\,\underline{\eta}(\theta) & \text{if } \Delta < 0, \\ \overline{\eta}(\theta,\Delta) & \text{if } \Delta > 0, \end{cases} \tag{S3}$$

where

a) $\underline{\eta}(\theta) > 0$ is bounded and independent of m ("cheap understatements", which reflect a decreasing but constant cost for controllers who understate their valuations).
b) $\overline{\eta}(\theta,\Delta)$ is continuous, strictly *increasing* and strictly *convex* in $\Delta > 0$, with the requirement that the two branches meet smoothly at the truth, i.e. $\overline{\eta}(\theta,0) = \underline{\eta}(\theta)$ ("costly talk-up", which reflect an increasing cost for controllers who overstate their valuations).

Type-Gradient: Higher-quality controllers face (weakly) lower marginal signalling cost:

$$C_\theta(m,\theta) \;\leq\; 0 \quad \text{for every } (m,\theta). \tag{S4}$$

which formalizes the idea that, for any given report m, controllers overseeing higher-quality projects incur lower costs when reporting their true valuations relative to lower-quality controllers.

Tail Growth: For each $\theta \in [\underline{\theta}_0, \bar{\theta}_0]$, the marginal "talk-up" cost is eventually prohibitive:

$$\lim_{\Delta\to\infty} C(\theta+\Delta,\theta) \;=\; \infty.$$

Equivalently, with $\overline{\eta}(\theta,t) := \partial_m C(\theta+t,\theta)$, $\int_0^\Delta \overline{\eta}(\theta,t)\,dt \to \infty$ as $\Delta \to \infty$.

2.1.3 Controlling Shareholder's Utility Function

Given the primitives in Sections 2.1.1 and 2.1.2, and the voting cut-off in (V2), define the rule-specific approval indicator

$$\mathbf{1}\{\text{approve}(m)\} \;:=\; \mathbf{1}\Big\{(1-\alpha)\,\mu(m) \;-\; \beta(m) \;\geq\; \kappa^{\text{rule}}(\lambda)\Big\} \;=\; \mathbf{1}\{\mu(m) \;\geq\; u\},$$

where u is the belief cut-off from (V3). The controlling shareholder chooses a message anticipating minority voting:

$$\max_{m}\; U_{CS}(m,\theta) \;=\; \mathbf{1}\{\text{approve}(m)\}\cdot\Big(\alpha\,\theta + B(\theta)\Big) \;-\; C(m,\theta). \tag{C1}$$

2.2 Minority Shareholders

2.2.1 Minority Shareholder Voting and Posterior Beliefs

As discussed in Section 2.1.1, a large but finite number N of symmetric minority shareholders collectively hold the residual $1-\alpha$ of the project's valuation, θ. To simplify notation, I define the variables s and λ:

$$s \;=\; \frac{1-\alpha}{N}, \qquad \lambda \;:=\; \frac{s}{1-\alpha} \;=\; \frac{1}{N} \;\in\; (0,1).$$

Hence each minority shareholder owns the fraction λ of the residual stake and, correspondingly, is exposed to a λ-proportion of any private benefits extracted by the controlling shareholder. I assume $\sum_{n=1}^{N} \lambda\, B(\theta) \;=\; B(\theta)$, implying that the controlling shareholder's extraction of private benefits is zero-sum. Except where noted, I will henceforth treat the case of a single representative minority shareholder, without loss of generality.

2.2.2 Beliefs and Updating

For each minority shareholder, a *belief system* specifies, for each publicly observed message $m \in \mathbb{R}$, a posterior distribution $\rho(\cdot \mid m) \in \Delta([\underline{\theta}_0, \overline{\theta}_0])$ over types. On the equilibrium path, beliefs are obtained by Bayes' rule from the prior F and the sender's strategy σ; off the equilibrium path we adopt the Grossman–Perry *credible-beliefs* refinement (Appendix A.2): upon observing an unexpected message, minorities place probability 1 on the set of types that (weakly) benefit most from that deviation (Grossman and Perry, 1986).

We summarize posteriors by their expectations

$$\mu(m) \;:=\; \mathbb{E}[\theta \mid m], \qquad \beta(m) \;:=\; \mathbb{E}[B(\theta) \mid m],$$

which by assumptions P1 and P3 always lie in $[\underline{\theta}_0, \overline{\theta}_0]$ and $[B(\underline{\theta}_0), B(\overline{\theta}_0)]$, respectively. In other words, $\mu(m)$ reflect's the minority shareholder's conditional expectation of the project's valuation

when they observe the message m. We assume that:

$\mu(m)$ is continuous, strictly increasing, and invertible; accordingly, μ^{-1} exists and is continuous. (M1)

2.2.3 Expected Utility from Approving (Interim Stage)

Following (Levit et al., 2024), I endow each minority investor i with a privately known bias b_i that perturbs her cut-off for approving the proposal. The bias is normalised as

$$b_i \sim \text{Uniform}\,[-1,\ 1], \qquad \mathbb{E}[b_i] = 0, \tag{M2}$$

independent of θ and distributed across investors. Using the posterior summaries from Section 2.2.2, investor i's interim utility upon approving is

$$\mathbb{E}[u_i \mid m] = \lambda\big[(1-\alpha)\,\mu(m) - \beta(m)\big] + b_i, \tag{M3}$$

with $G(b) = (b+1)/2$ for $b \sim U[-1,1]$.

2.2.4 Yes/No rule and Pivotal bias

I define

$$t_\lambda(m) := -\lambda\big[(1-\alpha)\mu(m) - \beta(m)\big].$$

Thus, a minority shareholder i votes "Yes" if and only if $b_i \geq t_\lambda(m)$. Accordingly, the fraction of "Yes" votes is

$$1 - G\big(t_\lambda(m)\big) = \frac{1 - t_\lambda(m)}{2}, \tag{M4}$$

given the distribution of $b_i \sim U[-1,1]$, for $t_\lambda(m) \in [-1,1]$ ($t_\lambda(m)$ saturates at 0 or 1 outside).

2.3 Voting Rules

2.3.1 Definition

I consider two types of Voting Rules.

1. **Majority of the Minority (MoM).** A proposal is implemented if and only if strictly more than 0.5 of the $(1-\alpha)$ minority shares vote "Yes."

2. **Super-majority (SM).** Fix a quota $S \in (0.5, 1)$. Implementation requires at least a fraction S of *all* shares in favour. Equivalently, the fraction of minority votes needed is

$$\pi \;=\; \frac{S-\alpha}{1-\alpha}. \tag{V1}$$

An SM rule is defined to be *lenient* when $\pi < 0.5$ and *stringent* when $\pi > 0.5$.

With these primitives fixed, all subsequent analysis derives equilibrium signalling and voting outcomes and compares welfare across the two rule families.

2.3.2 Cut-offs under MoM and super-majority S

For the MoM rule, the cutoff point is defined by the minority shareholder who is indifferent between voting "Yes" and "No" for a given proposal, conditional on the message m sent by the CS. As such, given expressions M4 and V1, under the MoM rule, shareholders vote "Yes" iff:

$$\frac{1-t_\lambda(m)}{2} \;\geq\; \tfrac{1}{2} \;\Longrightarrow\; t_\lambda(m) \leq 0 \;\Longrightarrow\; (1-\alpha)\,\mu(m) - \beta(m) \;\geq\; 0.$$

$$(1-\alpha)\,\mu(m) - \beta(m) \;\geq\; 0.$$

Similarly, under the SM rule, let $\pi = (S-\alpha)/(1-\alpha)$. Requiring $(1-t_\lambda(m))/2 \geq \pi$ as per expressions M4 and V1 gives the condition where shareholders vote "Yes" iff:

$$t_\lambda(m) \leq 1-2\pi \;\Longrightarrow\; (1-\alpha)\,\mu(m) - \beta(m) \;\geq\; -\frac{1-2\pi}{\lambda}.$$

$$(1-\alpha)\,\mu(m) - \beta(m) \;\geq\; -\frac{1-2\pi}{\lambda}.$$

We can then define the rule-specific cutoff

$$\underline{\kappa}^{(\lambda)}_{\text{rule}} := -\frac{k_{\text{rule}}}{\lambda}, \quad k_{\text{MoM}} = 0, \quad k_{\text{SM}} = 1-2\pi,$$

so that in both cases minorities approve iff:

$$(1-\alpha)\,\mu(m) - \beta(m) \;\geq\; \underline{\kappa}^{(\lambda)}_{\text{rule}}. \tag{V2}$$

2.3.3 Belief cut-off, minimal persuasive message, and strategy notation

Fix a voting rule and exposure λ. Let $\kappa_{\text{rule}}^{(\lambda)}$ be the constant in (V2). Define the minimal persuasive message which would pass the proposal given a voting rule:

$$m_{\text{req}} := \inf\left\{ m \in \mathbb{R} \;:\; (1-\alpha)\mu(m) - \beta(m) \;\geq\; \kappa_{\text{rule}}^{(\lambda)} \right\}, \qquad u := \mu(m_{\text{req}}). \tag{V3}$$

By (M1), μ is strictly increasing and continuous, so m_{req} exists and is unique with $m_{\text{req}} = \mu^{-1}(u)$. Thus, u is the conditional expected valuation of the pivotal shareholder who induces the proposal to pass.

Standing assumption (interior cut-off): Unless stated otherwise, $u \in (\theta_0, \bar{\theta}_0)$; the degenerate cases $u \notin (\theta_0, \bar{\theta}_0)$ (all-Yes/all-No) are collected in Corollary 1 (Appendix).

Strategy notation: The controller's message strategy depends on type θ and on the belief environment through u, such that $m = \sigma_u(\theta)$, or $m(\theta; u)$ when the u-dependence matters.

2.3.4 Vanishing Exposure as $\lambda \to 0$

Under MoM, $k_{\text{MoM}} = 0$ so the cutoff is always zero, so the shareholder dispersion (λ) has *no* effect. Under a lenient SM rule, $1 - 2\pi > 0$, so $\underline{\kappa}_{\text{SM}}^{(\lambda)} \to -\infty$ as $\lambda \to 0$. Hence, for sufficiently dispersed ownership, every posterior belief satisfies the inequality, and all projects are approved. In contrast, under a stringent SM rule, $1-2\pi < 0$; so the cut-off $\underline{\kappa}_{\text{SM}}^{(\lambda)} \to +\infty$ as $\lambda \to 0$. No posterior can clear the hurdle, so all projects are rejected in the atomistic limit. The following proposition formally establishes this intuition.

Proposition 1 (Limit approval outcomes as $\lambda \to 0$)**.** *Fix a voting rule.*

- *Under MoM ($\pi = \frac{1}{2}$) the approval set is unchanged by λ; some projects pass, some fail.*
- *Under a lenient SM ($\pi < \frac{1}{2}$) there exists $\bar{\lambda} > 0$ such that* $\Pr(\textit{approval} \mid \lambda < \bar{\lambda}) = 1$.
- *Under a stringent SM ($\pi > \frac{1}{2}$) there exists $\bar{\lambda} > 0$ such that* $\Pr(\textit{approval} \mid \lambda < \bar{\lambda}) = 0$.

Proof. By assumptions P1 and P3, $\mu(m) \in [\theta_0, \bar{\theta}_0]$ and, since B is increasing, $\beta(m) \in [B(\theta_0), B(\bar{\theta}_0)]$. Hence for every m,

$$(1-\alpha)\theta_0 - B(\bar{\theta}_0) \;\leq\; (1-\alpha)\mu(m) - \beta(m) \;\leq\; (1-\alpha)\bar{\theta}_0 - B(\theta_0). \tag{V4}$$

Under a MoM rule, $k_{\text{MoM}} = 0$ so $\underline{\kappa}_{\text{MoM}}^{(\lambda)} = 0$ is independent of λ.

Under a Lenient SM rule, $1 - 2\pi > 0$, so $\underline{\kappa}_{\text{SM}}^{(\lambda)} = -(1-2\pi)/\lambda \to -\infty$ as $\lambda \downarrow 0$. Choose $\bar{\lambda}$ so that $-(1-2\pi)/\bar{\lambda} \leq (1-\alpha)\theta_0 - B(\bar{\theta}_0)$; then for all $\lambda < \bar{\lambda}$, (V2) holds for every m by (V4), hence approval is certain.

Under a Stringent SM rule, $1 - 2\pi < 0$, so $\underline{\kappa}_{\mathrm{SM}}^{(\lambda)} = -(1-2\pi)/\lambda = (2\pi - 1)/\lambda \to +\infty$ as $\lambda \downarrow 0$. Pick $\bar{\lambda}$ such that $(2\pi - 1)/\bar{\lambda} \geq (1-\alpha)\bar{\theta}_0 - B(\theta_0)$; then for all $\lambda < \bar{\lambda}$ no posterior can clear (V2) by (V4), so rejection is certain. □

Intuitively, when ownership is highly dispersed ($\lambda \to 0$), a lenient SM rule becomes toothless because minority shareholders, bearing negligible personal exposure to diversion, approve virtually any proposal— reflecting the standard "free-riding" problem in corporate governance (Shleifer and Vishny (1986)). By contrast, a stringent SM rule becomes overly rigid, as no project can satisfy the inflated voting hurdle. The MoM rule lies precisely at the knife-edge: greater dispersion leaves its cut-off unchanged.

2.3.5 Closed-Form Cut-Offs with Linear Private Benefits

The belief cut-off u depends on the shape of the private-benefit function $B(\theta)$. To obtain a closed-form illustration, assume that the controller's private benefits are linear,

$$B(\theta) = \beta\,\theta, \qquad \beta \in [0, 1).$$

Given this specification, the conditional expectation in (V2) becomes

$$\beta(m) \;=\; \mathbb{E}\big[B(\theta) \mid m\big] \;=\; \beta\,\mu(m),$$

so the minority acceptance condition can be rewritten as

$$(1-\alpha)\,\mu(m) \;-\; \beta\,\mu(m) \;\geq\; \underline{\kappa}_{\mathrm{rule}}^{(\lambda)} \quad\Longleftrightarrow\quad (1-\alpha-\beta)\,\mu(m) \;\geq\; \underline{\kappa}_{\mathrm{rule}}^{(\lambda)}.$$

Because $1 - \alpha - \beta > 0$, this inequality implies the belief cut-off

$$\mu(m) \;\geq\; \underline{\theta}_{\mathrm{rule}}(\lambda) \;:=\; \frac{\underline{\kappa}_{\mathrm{rule}}^{(\lambda)}}{1-\alpha-\beta}$$

for each voting rule. Substituting the rule-specific constants yields

$$\underline{\theta}_{MoM} = \frac{0}{1-\alpha-\beta} = 0,$$

$$\underline{\theta}_{SM}(\lambda) = \frac{-(1-2\pi)/\lambda}{1-\alpha-\beta} = -\frac{1-2\pi}{\lambda\,(1-\alpha-\beta)}.$$

In a separating equilibrium, where each type θ sends a distinct message so that $\mu(m) = \theta$, the belief thresholds in (V2) translate directly into type thresholds. In a pooling equilibrium, minorities compare the pooled belief with the corresponding $\underline{\theta}_{\mathrm{rule}}(\lambda)$.

3 The Model: Signalling Incentives and Equilibrium Structure

We first prove four general lemmas that hold *for any voting rule* and *without* specifying functional forms for μ or C.

3.1 General lemmas

We reserve $u \in (\underline{\theta}_0, \bar{\theta}_0)$ for the rule-specific belief cut-off from (V2) and θ for types.[5] Define

$$m_{\text{req}}(u) := \mu^{-1}(u), \qquad \Lambda(\theta; u) := \alpha\,\theta + B(\theta) - C\big(m_{\text{req}}(u), \theta\big).$$

First, define the "cheapest effective lie" as the lowest-cost message required to induce approvals by the pivotal minority shareholders given their belief thresholds in V2:

$$m_{\text{req}}(u) \;:=\; \min\{\, m \,:\, \mu(m) \geq u \,\} = \mu^{-1}(u), \quad \text{strictly increasing in } u \text{ by (M1).} \tag{1}$$

Second, define the CS's net gain from lying as a "lying-profit function", $\Lambda(\theta; u)$. In other words, a "lying-profit function" determines the payoff to the CS from misrepresenting the true valuation of the project, θ, such that minority shareholders will accept the proposal in equilibrium. We define $\Lambda(\theta; u)$ as:

$$\Lambda(\theta; u) \;:=\; \alpha\,\theta + B(\theta) - C\big(m_{\text{req}}(u), \theta\big).$$

Lemma 1 (Sensitivity of Λ to the rule cut-off)**.** *For every (θ, u) with $\theta \geq u_0$*

$$\frac{\partial \Lambda}{\partial u}(\theta; u) \;=\; -\,C_m\big(m^{\text{req}}, \theta\big)\,\mu^{-1\prime}(u).$$

Because $\mu^{-1\prime}(u) > 0$ (M1), the sign is

$$\text{sign}\Big(\tfrac{\partial \Lambda}{\partial u}\Big) \;=\; -\,\text{sign}\Big(C_m(m^{\text{req}}, \theta)\Big) \;=\; \text{sign}\big(\theta - m^{\text{req}}\big).$$

In particular

$$\tfrac{\partial \Lambda}{\partial u} < 0 \;\; \textit{whenever } m^{\text{req}} > \theta \qquad \tfrac{\partial \Lambda}{\partial u} > 0 \;\; \textit{whenever } m^{\text{req}} < \theta$$

Proof. By definition $m^{\text{req}} = \mu^{-1}(u)$, so $\frac{\partial m^{\text{req}}}{\partial u} = \mu^{-1\prime}(u) > 0$ (M1). Differentiating $\Lambda(\theta; u) = \alpha\theta + B(\theta) - C(m^{\text{req}}, \theta)$ yields the displayed formula immediately. The sign conclusions use the sign convention for C_m in S3. By definition $m^{\text{req}}(\theta; u) = \mu^{-1}(u)$, hence

$$\frac{\partial m^{\text{req}}}{\partial u} = \mu^{-1\,\prime}(u) > 0 \qquad \text{(M1)}.$$

[5] We never write $u = \theta$.

Differentiate $\Lambda(\theta; u) = \alpha\theta + B(\theta) - C\big(m^{\text{req}}, \theta\big)$:

$$\frac{\partial \Lambda}{\partial u} = -\, C_m(\cdot)\, \frac{\partial m^{\text{req}}}{\partial u} \le 0 \qquad (\mu^{-1\prime} > 0 \text{ and } C_m \ge 0).$$

The sign is strict exactly when $m^{\text{req}}(\theta; u) > \theta$, because in that region $C_m > 0$ by (S3). □

Lemma 2 (Monotonicity of the zero-crossing in the cut–off)**.** *Fix a belief cut-off $u = u \in (u_0, \overline{\theta}_0]$. Recall*

$$m^{\text{req}}(u) \;:=\; \mu^{-1}(u), \qquad \Lambda(\theta; u) \;:=\; \alpha\theta + B(\theta) - C\big(m^{\text{req}}(u), \theta\big).$$

Assume $\Lambda(\cdot; u)$ is strictly single-crossing, so the equation $\Lambda(\theta; u) = 0$ has a unique solution $\theta^*(u) \in (u_0, u)$. Then $u \mapsto \theta^*(u)$ is continuously differentiable and

$$\frac{d\theta^*}{du} = -\frac{\Lambda_u\big(\theta^*(u); u\big)}{\Lambda_\theta\big(\theta^*(u); u\big)} \;>\; 0\,.$$

Hence the upper end of the pooling band θ^* moves strictly upward *when the voting cut-off u is tightened.*

Proof. Define the function $F(\theta, u) \;:=\; \Lambda(\theta; u)$. By definition of $\theta^*(u)$, we have $F\big(\theta^*(u), u\big) = 0$ for all u in the relevant domain.

Since $\Lambda(\theta; u) = \alpha\theta + B(\theta) - C\big(\mu^{-1}(u), \theta\big)$,

$$\frac{\partial F}{\partial u} = \frac{\partial \Lambda}{\partial u} = -\frac{\partial C}{\partial m}\big(\mu^{-1}(u), \theta\big) \cdot \frac{d\mu^{-1}}{du}(u).$$

By S3, $C_m(m, \theta) < 0$ when $m \le \theta$ and $C_m(m, \theta) > 0$ when $m > \theta$. Since $m^{\text{req}}(u) > \theta$ at the root, $C_m\big(\mu^{-1}(u), \theta\big) > 0$; together with $\mu^{-1\prime}(u) > 0$ from M1 this implies $\Lambda_u < 0$.

We next differentiate with respect to θ:

$$\frac{\partial F}{\partial \theta} = \alpha + B'(\theta) \;-\; C_\theta\big(\mu^{-1}(u), \theta\big).$$

Assumption S4 gives $C_\theta \le 0$, while P3 gives $B'(\theta) > 0$, hence

$$\Lambda_\theta = \frac{\partial F}{\partial \theta} \;\ge\; \alpha + B'(\theta) \;>\; 0.$$

Since $F\big(\theta^*(u), u\big) = 0$ defines θ^* implicitly as a function of u, and since F is continuously differentiable in both arguments (by continuity of C, B, and μ^{-1}), $\frac{\partial F}{\partial \theta}\big(\theta^*(u), u\big) \ne 0$ (by the single-crossing

property), the Implicit Function Theorem guarantees that $\theta^*(u)$ is differentiable with

$$\frac{d\theta^*}{du} = -\frac{\frac{\partial F}{\partial u}\big(\theta^*(u), u\big)}{\frac{\partial F}{\partial \theta}\big(\theta^*(u), u\big)} = -\frac{\Lambda_u\big(\theta^*(u); u\big)}{\Lambda_\theta\big(\theta^*(u); u\big)}.$$

Since $\Lambda_u < 0$ (as established earlier) and $\Lambda_\theta > 0$ (single-crossing), we have $\frac{d\theta^*}{du} > 0$. Therefore, θ^* is strictly increasing in u. □

Assumption 1 (Density–weighted boundary responsiveness (symmetric DWBR))**.** *Maintain Assumptions P1–P3, S1–S4, and M1. Let u be any interior belief cut-off (Def. V3) for which partial pooling obtains, with lower boundary $\theta^*(u)$. Write the pooling mass as $P(u) := F(u) - F(\theta^*(u))$.*

(Lenient side; $u \in [u_{\mathrm{SM}}(\pi), 0]$). *Along any interval of interior lenient cut-offs,*

$$\theta^{*\prime}(u) \;\geq\; \frac{f(u)}{f(\theta^*(u))}$$

Equivalently (by Lemma 2),

$$C_m\big(m^{\mathrm{req}}(u), \theta^*(u)\big)\, \mu^{-1\prime}(u) \;\geq\; \frac{f(u)}{f(\theta^*(u))}\Big[\alpha + B'\big(\theta^*(u)\big) - C_\theta\big(m^{\mathrm{req}}(u), \theta^*(u)\big)\Big].$$

(Stringent side; $u \in [0, \bar{u}]$ while pooling survives). *Along any interval of interior stringent cut-offs,*

$$\theta^{*\prime}(u) \;\leq\; \frac{f(u)}{f(\theta^*(u))}$$

Equivalently (by Lemma 2),

$$C_m\big(m^{\mathrm{req}}(u), \theta^*(u)\big)\, \mu^{-1\prime}(u) \;\leq\; \frac{f(u)}{f(\theta^*(u))}\Big[\alpha + B'\big(\theta^*(u)\big) - C_\theta\big(m^{\mathrm{req}}(u), \theta^*(u)\big)\Big].$$

Usage. *Differentiating $P(u) = F(u) - F(\theta^*(u))$ gives*

$$P'(u) = f(u) - f\big(\theta^*(u)\big)\, \theta^{*\prime}(u).$$

Hence the lenient inequality implies $P'(u) \leq 0$ on $[u_{\mathrm{SM}}(\pi), 0]$, while the stringent inequality implies $P'(u) \geq 0$ on $[0, \bar{u}]$ until the pooling region vanishes at the critical quota.

Lemma 3 (Pooling test)**.** *Let*

$$\Psi(u) \;:=\; \Lambda\big(\theta = \underline{\theta}_0\,;\, u\big),$$

i.e. the lying–profit function evaluated at the lowest quality in the support, $\theta = \underline{\theta}_0$. Then Ψ is continuous and weakly *decreasing in u; the inequality is strict whenever $m^{\mathrm{req}}(u) > \underline{\theta}_0$. Therefore,*

$$\Pr(\textit{pooling}) > 0 \quad\Longleftrightarrow\quad \Psi(u) < 0.$$

Proof. Because $\Lambda(\theta; u)$ is continuous in both arguments, its restriction to $\theta = \underline{\theta}_0$ is also continuous, so Ψ is continuous. Lemma 1 shows $\partial\Lambda/\partial u \leq 0$ at every θ, with strict inequality whenever the effective lie is upward; in particular

$$\Psi'(u) \;=\; \frac{\partial\Lambda}{\partial u}(\underline{\theta}_0; u) \;=\; -\,C_m\big(m^{\text{req}}, \underline{\theta}_0\big)\,\mu^{-1\prime}(u).$$

Since $u = u > \underline{\theta}_0$, we have $m^{\text{req}} = \mu^{-1}(u) > \underline{\theta}_0$, hence $C_m > 0$ by S3; with $\mu^{-1\prime} > 0$ the derivative is strictly negative.

Assume first that $\Psi(u) \geq 0$. Since $\Lambda(\theta; u)$ is strictly increasing in θ (single-crossing), the inequality $\Lambda(\underline{\theta}_0; u) \geq 0$ implies $\Lambda(\theta; u) \geq 0$ for every $\theta \geq \underline{\theta}_0$. The zero-crossing $\theta^*(u)$ therefore satisfies $\theta^* \leq \underline{\theta}_0$, so the interval $[\underline{\theta}_0, \theta^*)$ is empty and the probability of pooling, $F(\theta^*) - F(\underline{\theta}_0)$, equals 0.

Conversely, if $\Psi(u) < 0$ then $\Lambda(\underline{\theta}_0; u) < 0$. With $\Lambda(\cdot; u)$ strictly increasing, the function must cross zero at some $\theta^* > \underline{\theta}_0$, leaving a non-empty interval $[\underline{\theta}_0, \theta^*)$ on which $\Lambda < 0$; hence $F(\theta^*) - F(\underline{\theta}_0) > 0$.

Thus Pr(pooling) > 0 exactly when $\Psi(u) < 0$. □

Lemma 4 (Basic shape)**.** *$\Lambda(\theta; u)$ is continuous, strictly increasing in θ, and weakly decreasing in u (strictly decreasing whenever $m^{\text{req}}(\theta; u) > \theta$).*

Proof. Monotonicity in θ follows from $B'(\theta) > 0$ and single-crossing (log-concave f, convex C). Monotonicity in u is Lemma 1. □

3.2 General Existence and Uniqueness

As pointed out by Burkart et al. (2024), signaling problems often induce multiple equilibria (Shleifer and Vishny (1986)). To obtain a unique equilibrium outcome, I follow Burkart et al. (2024) in imposing the assumption that off-equilibrium beliefs satisfy the credible beliefs criterion of Grossman and Perry (1986), together with the log-concave density function of the CS's type as per Assumption P2. As per Bagnoli and Bergstrom (2005), this is a common assumption in signaling games with threshold strategies. In what follows, I define the existence of the unique equilibrum in these games— generalized to both voting rules.

3.2.0.1 Scope.

By the interior cut-off assumption, we take $u \in (\theta_0, \bar{\theta}_0)$. The degenerate edge cases are collected in Corollary 1.

Proposition 2 (Perfect–Sequential Equilibrium for a fixed belief cut-off)**.** *Fix any rule that induces the belief cut-off $u \in (\underline{\theta}_0, \bar{\theta}_0]$. Define $m_{\text{req}}(u) := \mu^{-1}(u)$ and $\Lambda(\theta; u) := \alpha\theta + B(\theta) - C(m_{\text{req}}(u), \theta)$ as per Section 3.1. Let $\theta^*(u) := \sup\{\theta \in [\underline{\theta}_0, u) : \Lambda(\theta; u) < 0\}$. Minorities implement iff $\mu(m) \geq u$ (equivalently, iff $(1-\alpha)\mu(m) - \beta(m) \geq \underline{\kappa}^{(\lambda)}_{rule}$ by (V2)). Assume GP–credible beliefs off path.*

(i) There exists a PSE with sender strategy

$$\sigma(\theta) = \begin{cases} \theta, & \theta < \theta^*(u) \quad \textit{(truth; rejected)}, \\ m_{\text{req}}(u), & \theta \in [\theta^*(u), u) \quad \textit{(cheapest lie; approved)}, \\ \theta, & \theta \geq u \quad \textit{(truth; approved)}. \end{cases}$$

(ii) For the same cut-off u, every PSE yields the same implement/reject allocation: the project is implemented iff $\theta \geq \theta^*(u)$.

(iii) The equilibrium is partially pooling iff $\theta^*(u) > \underline{\theta}_0$; otherwise it is fully separating.

Proof. Preliminaries (voting, feasibility, and monotonicity). By (M1), μ is continuous, strictly increasing, and invertible, so $m_{\text{req}}(u)$ exists and is unique. Given (M3) and $b_i \sim U[-1, 1]$, investor i votes "Yes" iff $b_i \geq t_\lambda(m)$, which yields a Yes-vote fraction $\big[1 - t_\lambda(m)\big]/2$ (M4). Combining with the rule's quota gives the common form (V2): *approval iff $\mu(m) \geq u$*. Hence $m_{\text{req}}(u) = \mu^{-1}(u)$ is indeed the smallest message that passes. Define $\Lambda(\theta; u) = \alpha\theta + B(\theta) - C\big(m_{\text{req}}(u), \theta\big)$. By assumptions S3, S4 and the log-concavity assumption (P2), (a) $\theta \mapsto \Lambda(\theta; u)$ is continuous and strictly increasing, (b) $u \mapsto \Lambda(\theta; u)$ is (weakly) decreasing with strict inequality whenever $m_{\text{req}}(u) > \theta$ (Lemmas 1 and 4 in Section 3.1).

(i) Existence. We verify sequential rationality type-by-type and then the receivers' best responses.

Step 1 (low types $\theta < \theta^*(u)$). By definition of $\theta^*(u)$, $\Lambda(\theta; u) < 0$. If a low type deviates upward to the cheapest approved message $m_{\text{req}}(u)$, her payoff is

$$\alpha\theta + B(\theta) - C\big(m_{\text{req}}(u), \theta\big) = \Lambda(\theta; u) < 0,$$

which is strictly below the truthful-rejection payoff 0. Any *other* approved message has $m > m_{\text{req}}(u)$ and—because for $m > \theta$ the marginal cost $C_m > 0$ and strictly increases in the talk-up wedge (S3)—incurs strictly higher cost, hence is *a fortiori* worse. Any under-report $m < \theta$ is rejected (since μ is increasing, $\mu(m) < \mu(\theta) = \theta < u$), and by S2–S3 has strictly higher cost than truth ($C(\theta, \theta) = 0$ and for $m < \theta$, $C_m = -\eta(\theta) < 0$ implies $C(m, \theta) > 0$). Therefore truth maximizes payoff for all $\theta < \theta^*(u)$.

Step 2 (persuaders $\theta \in [\theta^*(u), u)$). Truth would be rejected ($\mu(\theta) = \theta < u$), yielding payoff 0. Sending $m_{\text{req}}(u)$ secures approval with net payoff $\Lambda(\theta; u) \geq 0$. Among approved messages, $m_{\text{req}}(u)$ minimizes cost because for $m > \theta$ the marginal cost is strictly positive and strictly increasing in m (S3), so any $m > m_{\text{req}}(u)$ is strictly worse; and any $m \leq \theta$ fails (since $\mu(m) \leq \theta < u$). Hence

$m_{\text{req}}(u)$ is strictly optimal on this region.

Step 3 (high types $\theta \geq u$). Truth is approved at zero signaling cost (S2). Any upward deviation $m > \theta$ strictly raises cost (S3) but yields no incremental benefit; any downward deviation either (i) stays above the threshold ($\mu(m) \geq u$) yet incurs positive cost because $C_m < 0$ on $m < \theta$, or (ii) falls below u and is rejected. In both cases truth strictly dominates. Hence $\sigma(\theta) = \theta$ is optimal.

Step 4 (minority best responses and beliefs). Investor i's interim utilities are $U_i^{\text{Yes}}(m) = \lambda\big[(1-\alpha)\mu(m) - \beta(m)\big] + b_i$ and $U_i^{\text{No}}(m) = b_i$. Hence i votes Yes iff

$$(1-\alpha)\mu(m) - \beta(m) \;\geq\; \underline{\kappa}_{\text{rule}}^{(\lambda)},$$

which is the common cut-off form (V2). On-path truths $m = \theta$ with $\theta \geq u$ satisfy the inequality by definition of u; on-path pooling messages are below u and are rejected; and any off-path message receives GP-credible beliefs and fails the inequality. Thus the stated σ is a best response for minorities under the maintained rule.

All players are best-responding and beliefs satisfy Bayes' rule on path and GP off path; therefore the profile is a PSE.

(ii) Uniqueness of allocation. Let $(\hat{\sigma}, \widehat{\text{beliefs}})$ be any PSE with the *same* belief cut-off u as in part (i). Write $m_{\text{req}}(u) := \mu^{-1}(u)$ and $\Lambda(\theta; u) = \alpha\theta + B(\theta) - C\big(m_{\text{req}}(u), \theta\big)$. Let $\theta^*(u)$ be the unique zero of $\Lambda(\cdot; u)$ on $(\underline{\theta}_0, u)$ (Lemma 4). We prove that *every* PSE with cut-off u induces the same allocation:

$$\text{the project is implemented iff } \;\theta \geq \theta^*(u).$$

Claim A (No type below θ^* can be implemented). Fix $\theta < \theta^*(u)$ and suppose, toward a contradiction, that $\hat{\sigma}(\theta)$ yields approval under $\widehat{\text{beliefs}}$, i.e. $\mu(\hat{\sigma}(\theta)) \geq u$. Among all approved messages, the least costly for type θ is $m_{\text{req}}(u)$, because for $m > \theta$ the marginal signaling cost is strictly positive and increasing (S3), and μ is increasing so any approved message must lie weakly above $m_{\text{req}}(u)$. Hence

$$\alpha\theta + B(\theta) - C\big(\hat{\sigma}(\theta), \theta\big) \;\leq\; \alpha\theta + B(\theta) - C\big(m_{\text{req}}(u), \theta\big) \;=\; \Lambda(\theta; u) \;<\; 0.$$

By contrast, rejecting via any message with $\mu(m) < u$ yields payoff at most $-C(m, \theta)$, which is *weakly* maximized at the truth $m = \theta$ (S2 and $C_m < 0$ for $m < \theta$ under S3), delivering 0. Thus truthful rejection strictly dominates approval for $\theta < \theta^*(u)$, contradicting optimality. Therefore, no type below $\theta^*(u)$ can be implemented in any PSE with cut-off u.

Claim B (Every type above θ^* must be implemented). Fix $\theta > \theta^*(u)$ and suppose $\hat{\sigma}(\theta)$ leads to rejection, i.e. $\mu(\hat{\sigma}(\theta)) < u$. Deviating to $m_{\text{req}}(u)$ guarantees approval and yields

$$\alpha\theta + B(\theta) - C\big(m_{\text{req}}(u), \theta\big) = \Lambda(\theta; u) \;>\; 0,$$

which strictly improves upon any rejected payoff (at most 0 from truth). Hence rejection cannot be optimal. If instead $\hat{\sigma}(\theta)$ yields approval via some $m \neq \theta$, then for $\theta \geq u$ truth is approved at zero cost and strictly dominates any other approved message (S2–S3). For $\theta \in (\theta^*(u), u)$, $m_{\text{req}}(u)$

is the least-cost approved message; any approved m with $m > m_{\text{req}}(u)$ yields strictly lower payoff (S3). Thus types $\theta > \theta^*(u)$ are implemented in any PSE (possibly by truth when $\theta \geq u$, or via $m_{\text{req}}(u)$ when $\theta \in (\theta^*(u), u)$).

Boundary type. At $\theta = \theta^*(u)$ we have $\Lambda(\theta^*; u) = 0$ by definition. This type is indifferent between truthful rejection (payoff 0) and sending $m_{\text{req}}(u)$ to obtain approval (also payoff 0). Our convention includes $\theta^*(u)$ on the implemented side ("$\geq \theta^*$"); any tie-break that places $\theta^*(u)$ on either side leaves the allocation of all *other* types unchanged.

Combining Claims A and B delivers the result: in any PSE with cut-off u, implementation occurs iff $\theta \geq \theta^*(u)$. □

(iii) Pooling vs. separation. Evaluate Λ at the bottom of the support: $\Psi(u) := \Lambda(\underline{\theta}_0; u)$. Because $\Lambda(\cdot; u)$ is strictly increasing in θ (Lemma 4), two cases obtain:

Case 1 ($\Psi(u) < 0$): then the unique zero satisfies $\theta^*(u) > \underline{\theta}_0$, so the pooling set $[\underline{\theta}_0, \theta^*(u))$ has strictly positive probability $F(\theta^*(u)) - F(\underline{\theta}_0) > 0$.

Case 2 ($\Psi(u) \geq 0$): then $\theta^*(u) = \underline{\theta}_0$ and the pooling set is empty, yielding full separation. This equivalence is exactly Lemma 3. □

In general, the economic intuition of Proposition 2 reflects the general problem of adverse selection where signaling is costly. In particular, because sending signals are costly, controllers with very low θ will often find it profitable to remain silent—paying the requisite cost to induce approval of the project yields strictly less than simply saying nothing and getting zero. On the other hand, the nature of the CS's utility function (C1) suggests that the net benefits from approval rise linearly with θ, while the convex costs from upward misreporting imply that θ does not rise as rapidly for controllers with very high types. Accordingly, these controllers will always refuse to stay in a rejected pool. As we will demonstrate later, this may generalize to the possibility that all controllers may refuse to stay in a rejected pool.

Proposition 3 (Comparative statics of the pooling probability (transferable B))**.** *Fix a belief cut-off $u \leq 0$ and let $\theta^*(u) < 0$ be the unique root of $\Lambda(\theta; u) = 0$ on $(\underline{\theta}_0, u)$, where*

$$\Lambda(\theta; u) \;:=\; \alpha\,\theta + B(\theta) - C\big(m_{\text{req}}(u), \theta\big), \quad m_{\text{req}}(u) = \mu^{-1}(u).$$

Define the pooling probability *as the mass of types who send $m_{\text{req}}(u)$:*

$$P(u) \;:=\; F(u) \;-\; F\big(\theta^*(u)\big).$$

Consider the following exogenous parameter changes (holding the voting rule, and hence u, fixed):

(a) Upward quality shift: *an additive shift $\Delta > 0$ in fundamentals so that the CS's lying-profit becomes $\Lambda_\Delta(\theta; u) := \alpha(\theta + \Delta) + B(\theta + \Delta) - C(m_{\text{req}}(u), \theta + \Delta)$;*
(b) Higher ownership: *$\alpha \mapsto \alpha + \Delta\alpha$, with $\Delta\alpha > 0$;*
(c) Upward level shift in private benefits: *$B(\theta) \mapsto B(\theta) + \Delta B$, with $\Delta B > 0$.*

Then:

$$\begin{aligned} \textit{(a) Upward quality } (\Delta > 0) &: \quad P \textit{ strictly} \text{ increases,} \\ \textit{(b) Higher ownership } (\Delta\alpha > 0) &: \quad P \textit{ strictly} \text{ decreases,} \\ \textit{(c) Higher private benefits } (\Delta B > 0) &: \quad P \textit{ strictly} \text{ increases.} \end{aligned}$$

Proof. All derivatives are evaluated at the root $\theta^*(u) < 0$. By Lemma 4, $\Lambda_\theta(\theta; u) > 0$ (strict single-crossing), so by the Implicit Function Theorem

$$\frac{d\theta^*}{d\,\text{param}} = -\frac{\Lambda_{\text{param}}}{\Lambda_\theta} \quad \text{at } \theta = \theta^*(u).$$

Since $P(u) = F(u) - F(\theta^*(u))$ with fixed u, we have

$$\frac{dP}{d\,\text{param}} = -f\big(\theta^*(u)\big)\,\frac{d\theta^*}{d\,\text{param}},$$

and the density satisfies $f > 0$ on the support (Assumption P2). We now compute Λ_{param} in each case.

(a) Upward quality $\Delta > 0$. By construction, Δ enters Λ *only* through $\theta + \Delta$, hence $\Lambda_\Delta(\theta; u) = \Lambda_\theta(\theta; u) > 0$ (Assumptions P3 and S4 ensure $\alpha + B'(\cdot) - C_\theta(\cdot, \cdot) > 0$). Therefore

$$\frac{d\theta^*}{d\Delta} = -\frac{\Lambda_\Delta}{\Lambda_\theta} = -1 < 0,$$

and thus $\dfrac{dP}{d\Delta} = -f(\theta^*)\,\dfrac{d\theta^*}{d\Delta} = f(\theta^*) > 0$. Hence P strictly increases.

(b) Ownership α. Here $\Lambda_\alpha(\theta; u) = \theta$, so at the root $\Lambda_\alpha = \theta^*(u) < 0$, whence

$$\frac{d\theta^*}{d\alpha} = -\frac{\Lambda_\alpha}{\Lambda_\theta} > 0 \quad \Longrightarrow \quad \frac{dP}{d\alpha} = -f(\theta^*)\,\frac{d\theta^*}{d\alpha} < 0.$$

Thus P strictly decreases.

(c) Level shift in B. A constant increase adds one-for-one to Λ, i.e. $\Lambda_{\Delta B} = +1$. Hence

$$\frac{d\theta^*}{d(\Delta B)} = -\frac{1}{\Lambda_\theta} < 0 \quad \Longrightarrow \quad \frac{dP}{d(\Delta B)} = -f(\theta^*)\,\frac{d\theta^*}{d(\Delta B)} > 0.$$

All inequalities are strict because $f(\theta^*) > 0$ (P2) and $\Lambda_\theta > 0$ (Lemma 4). □

The intuition of Proposition 3 is simple. When projects are stronger across the board ($\Delta > 0$), approval is more valuable for every controller and, in economic terms, it becomes easier for marginal types to justify "talking up" just enough to clear the same "lying" hurdle. More of the previously borderline (slightly negative) projects now find it worthwhile to send the minimal

persuasive message. The marginal type who is just indifferent shifts downward, so the pooling band widens and the pooling probability rises.

Owning a larger equity stake ($\alpha \uparrow$) makes the controller internalize more of the project's cash flows. For the low or middling projects that were tempted to exaggerate, this cuts the prize from getting the deal through—because they now bear more of the downside. With a smaller prize, fewer marginal types are willing to pay the cost of exaggeration. The marginal indifferent type moves upward, so the pooling band shrinks and the pooling probability falls.

A uniform increase in private benefits ($B \uparrow$) is an extra gain that arrives only if the deal is approved; while it doesn't make the message any more convincing under a fixed threshold, it raises the payoffs of controllers conditional on approval of the project. With a bigger prize for approval, more types are willing to exaggerate just enough to pass. The marginal indifferent type moves downward, so the pooling band widens and the pooling probability rises.

4 Pooling Regions under Different Voting Regimes

4.1 Majority-of-Minority versus a Lenient Super-Majority

The following propositions demonstrate a comparison between MoM and Lenient SM, where the requisite quota, π , is less than 0.5. We first show that both rules induce partial pooling, before showing that the MoM induces relatively less pooling than the lenient SM rule.

Proposition 4 (Both MoM and any lenient SM induce partial pooling)**.** *Maintain Assumptions in Section 2 with $\underline{\theta}_0 < 0 < \overline{\theta}_0$ and the normalisation $B(\underline{\theta}_0) = 0$. Let $u \leq 0$ be the belief cut-off generated by a voting rule ($u = 0$ for MoM and $u_{\mathrm{SM}}(\pi) = -(1-2\pi)/\lambda < 0$ for any lenient quota $\pi < \frac{1}{2}$).*

Define

$$m^{\mathrm{req}}(u) := \mu^{-1}(u), \qquad \Lambda(\theta; u) := \alpha\,\theta + B(\theta) - C\big(m^{\mathrm{req}}(u), \theta\big),$$

and the zero of the lying–profit curve

$$\theta^*(u) := \sup\Big\{\, \theta \in [\underline{\theta}_0, u) : \ \Lambda(\theta; u) < 0 \Big\}.$$

Then, for every $u \leq 0$,

$$\underline{\theta}_0 < \theta^*(u) < u \quad \Longrightarrow \quad F\big(\theta^*(u)\big) - F(\underline{\theta}_0) > 0,$$

so the unique Perfect-Sequential Equilibrium is partially pooling*: types $\theta < \theta^*(u)$ tell the truth and are rejected, whereas all $\theta \in [\theta^*(u), u)$ pool on $m_{\mathrm{req}}(u)$ and are approved; types $\theta \geq u$ separate truthfully and are approved.*

Proof. First, we evaluate the sign of Λ at $\underline{\theta}_0$. Since $u \le 0$ and μ is strictly increasing,

$$m^{\text{req}}(u) = \mu^{-1}(u) > \underline{\theta}_0.$$

Thus at $\theta = \underline{\theta}_0$ the cost argument satisfies $m^{\text{req}}(u) > \theta$, so $C\big(m^{\text{req}}(u), \underline{\theta}_0\big) > 0$. Using $B(\underline{\theta}_0) = 0$ and $\alpha\, \underline{\theta}_0 < 0$,

$$\Lambda(\underline{\theta}_0; u) = \alpha\, \underline{\theta}_0 - C\big(m^{\text{req}}(u), \underline{\theta}_0\big) < 0. \tag{P1}$$

Next, we show that there exists a unique zero strictly between $\underline{\theta}_0$ and u. By Lemma 4, $\theta \mapsto \Lambda(\theta; u)$ is continuous and strictly increasing. Since $\Lambda(\underline{\theta}_0; u) < 0$ by (P1), and at $\theta = u$

$$\Lambda(u; u) = \alpha\, u + B(u) - C(u, u) = \alpha\, u + B(u) > 0,$$

there is a unique zero $\theta^*(u) \in (\underline{\theta}_0, u)$. Finally, we show that there is a positive measure of pooling types. Because F is strictly increasing,

$$F\big(\theta^*(u)\big) - F(\underline{\theta}_0) > 0.$$

Proposition 2 (iii) then yields the stated partial-pooling equilibrium: all $\theta < \theta^*(u)$ pool, while $\theta \ge \theta^*(u)$ truthfully separate. □

Proposition 5 (A lenient SM yields a *larger pooling mass* than MoM)**.** *Maintain Assumptions P1–P3, S1–S4, M1, the zero-mode refinement of P2 (i.e., $0 \in \mathcal{M}(f)$), and the symmetric DWBR Assumption 1 on the lenient side. Let $u_{\text{MoM}} = 0$ be the MoM belief cut-off and, for each lenient quota $\pi \in (0, \frac{1}{2})$, let*

$$u_{\text{SM}}(\pi) \;:=\; -\frac{1 - 2\pi}{\lambda} \;\in\; (\underline{\theta}_0, 0).$$

For every $u \le 0$ define $m^{\text{req}}(u) := \mu^{-1}(u)$, the lying-profit function

$$\Lambda(\theta; u) := \alpha\, \theta + B(\theta) - C\big(m^{\text{req}}(u), \theta\big),$$

and the unique zero $\theta^*(u) := \sup\{\theta < u : \Lambda(\theta; u) < 0\}$ (Lemma 4). Let the pooling mass be $P(u) := F(\theta^*(u)) - F(\underline{\theta}_0)$.

Then, for every lenient quota $\pi \in (0, \frac{1}{2})$,

$$P\big(u_{\text{SM}}(\pi)\big) \;>\; P(0)$$

i.e., any lenient SM produces a strictly larger pooling probability than MoM.

Proof. Step 1 (Differentiating the pooling mass). On any interval of interior cut-offs u for which partial pooling obtains (Proposition 4), $\theta^*(u) \in (\underline{\theta}_0, u)$ and θ^* is differentiable by the implicit-function formula at the unique zero of $\Lambda(\cdot; u)$:

$$\theta^{*\prime}(u) \;=\; -\frac{\Lambda_u(\theta^*(u); u)}{\Lambda_\theta(\theta^*(u); u)} \;=\; \frac{C_m\big(m^{\text{req}}(u), \theta^*(u)\big)\, \mu^{-1\prime}(u)}{\alpha + B'\big(\theta^*(u)\big) - C_\theta\big(m^{\text{req}}(u), \theta^*(u)\big)},$$

using $\Lambda_u(\theta; u) = -C_m(m^{\text{req}}(u), \theta)\, \mu^{-1}{}'(u)$ and $\Lambda_\theta = \alpha + B'(\theta) - C_\theta(m^{\text{req}}(u), \theta)$. By (S3)–(S4) and (M1), the denominator is positive and $C_m > 0$ at the pooling boundary (since $m^{\text{req}}(u) > \theta^*(u)$ when $u < 0$), so $\theta^{*\prime}(u) > 0$.

Define $P(u) = F(\theta^*(u)) - F(\underline{\theta}_0)$. By the chain rule,

$$P'(u) \;=\; f\big(\theta^*(u)\big)\, \theta^{*\prime}(u).$$

Equivalently (and more useful below), subtract and add $f(u)$ to obtain

$$P'(u) \;=\; f(u) - \Big[f(u) - f\big(\theta^*(u)\big)\Big] - f\big(\theta^*(u)\big)\Big[1 - \theta^{*\prime}(u)\Big] \;=\; f(u) - f\big(\theta^*(u)\big)\, \theta^{*\prime}(u). \tag{1}$$

Step 2 (Shape of f under a zero-mode prior). By Assumption P2 with $0 \in \mathcal{M}(f)$, f is unimodal with a (possibly flat) mode at 0 and therefore weakly nondecreasing on $[\underline{\theta}_0, 0]$. Hence, for all $u \in [u_{\text{SM}}(\pi), 0]$ with $\theta^*(u) < u \le 0$,

$$f\big(u\big) \;\ge\; f\big(\theta^*(u)\big). \tag{2}$$

Step 3 (DWBR bound on the lenient side). Assumption 1 (lenient side) yields, for all interior lenient cut-offs $u \in [u_{\text{SM}}(\pi), 0]$,

$$\theta^{*\prime}(u) \;\ge\; \frac{f(u)}{f\big(\theta^*(u)\big)}. \tag{3}$$

Step 4 (Sign of $P'(u)$ on $[u_{\text{SM}}(\pi), 0]$). Combining (1) and (3),

$$P'(u) \;=\; f(u) - f\big(\theta^*(u)\big)\, \theta^{*\prime}(u) \;\le\; f(u) - f\big(\theta^*(u)\big)\, \frac{f(u)}{f\big(\theta^*(u)\big)} \;=\; 0,$$

with inequality strict on a set of positive measure whenever either (i) $f(u) > f(\theta^*(u))$ on a subset (i.e., the left shoulder is not everywhere flat), or (ii) the DWBR bound (3) holds strictly on a subset. These are mild nondegeneracy conditions.

Step 5 (Integrating from the lenient cut-off up to MoM). Integrate $P'(u) \le 0$ over $u \in [u_{\text{SM}}(\pi), 0]$:

$$P(0) - P\big(u_{\text{SM}}(\pi)\big) \;=\; \int_{u_{\text{SM}}(\pi)}^{0} P'(u)\, du \;<\; 0,$$

where strictness follows from Step 4. Rearranging gives $P\big(u_{\text{SM}}(\pi)\big) > P(0)$, which proves the claim. □

The intuition of Propositions 4 and 5 may be summarized as follows. For the lenient SM rule, the requisite threshold sits below the unconditional mean. In particular, any upward nudge in the posterior may be able to tip the vote (to approval), so low-quality controllers see a high marginal return to push through a small lie (i.e., a small deviation of m from true observed

θ. Because the signalling cost is convex, these controllers will not push the message very far, but—crucially—many low types still find it strictly profitable to misreport, so the pooling set is wide. On the other hand, for the MoM rule, the rule sits at the exact point where half of minority votes are needed. For a bad project, the required upward shift in beliefs is larger than under a lenient SM rule, so fewer low types find it worthwhile to pay the signalling cost. For a good project, the hurdle is still attainable without over-exaggerating, so high types continue to report truthfully. However, as I will demonstrate later, this logic has to be modified when we examine the stringent SM rule.

4.2 Majority-of-Minority versus a Stringent Super-Majority

The following propositions demonstrate a comparison between MoM and Stringent SM, where the requisite quota, π , is more than 0.5. We first show that the stringent SM may induce complete separation for a sufficiently high quota, $\bar{\pi}$, before showing that the MoM induces relatively less pooling than the stringent SM rule when pooling survives in the stringent SM.

Proposition 6 (Equilibrium under a super-majority with quota π)**.** *Maintain Assumptions P1–P3, S1–S4, M1, and the tail-growth assumption on talk-up costs (Assumption* **??***). For each quota* $\pi \in (0,1)$*, define the* interior belief cut-off $u_{\mathrm{SM}}(\pi) \in (\underline{\theta}_0, \overline{\theta}_0)$ *as the unique solution to*

$$(1-\alpha)\, u - \beta\big(\mu^{-1}(u)\big) \;=\; -\frac{1-2\pi}{\lambda}. \tag{SM-cut}$$

Let $m^{\mathrm{req}}(u) := \mu^{-1}(u)$ *and* $\Lambda(\theta; u) := \alpha\theta + B(\theta) - C\big(m^{\mathrm{req}}(u), \theta\big)$*, and define* $\theta^*(u) := \sup\{\theta < u : \Lambda(\theta; u) < 0\}$*.*

(i) Existence and monotonicity of $u_{\mathrm{SM}}(\pi)$***.*** *For every* $\pi \in (0,1)$ *there exists a unique* $u_{\mathrm{SM}}(\pi) \in (\underline{\theta}_0, \overline{\theta}_0)$ *solving (SM-cut). Moreover,* $\pi \mapsto u_{\mathrm{SM}}(\pi)$ *is strictly increasing on* $(0,1)$*.*

(ii) Partial pooling at and below MoM. *At* $\pi = \frac{1}{2}$ *and for every lenient quota* $\pi < \frac{1}{2}$*, the unique PSE is partially pooling:* $\underline{\theta}_0 < \theta^*\big(u_{\mathrm{SM}}(\pi)\big) < u_{\mathrm{SM}}(\pi) \le 0$*.*

(iii) Critical quota and full separation (tail-growth). *There exists at least one* $\bar{\pi} \in (\frac{1}{2}, 1]$ *such that* $\theta^*\big(u_{\mathrm{SM}}(\bar{\pi})\big) = u_{\mathrm{SM}}(\bar{\pi})$*. Hence, for every* $\pi \ge \bar{\pi}$*, the unique PSE is fully separating (no pooling). If, in addition,* $L(u) := u - \theta^*(u)$ *is strictly decreasing on* $[u_{\mathrm{SM}}(\frac{1}{2}), u_{\mathrm{SM}}(1)]$ *(e.g., under the DWBR/slope-dominance condition), then* $\bar{\pi}$ *is unique.*

Affine-B corollary. *If* $B(\theta) = b_0 + b_1\theta$ *with* $b_1 \in (0, 1-\alpha)$*, then* $\beta\big(\mu^{-1}(u)\big) = b_0 + b_1 u$*, and (SM-cut) yields*

$$u_{\mathrm{SM}}(\pi) = \frac{-\,b_0 - \frac{1-2\pi}{\lambda}}{1-\alpha-b_1}, \qquad \frac{d u_{\mathrm{SM}}}{d\pi} = \frac{2/\lambda}{1-\alpha-b_1} > 0.$$

Proof. (i) Existence and monotonicity. Define $H(u,\pi) := (1-\alpha)\, u - \beta\big(\mu^{-1}(u)\big) + \frac{1-2\pi}{\lambda}$. By (M1), μ^{-1} is continuous and strictly increasing. By (P3), B is C^1 with $0 < B'(\theta) < 1-\alpha$; combined with

monotone-likelihood-ratio/single-crossing of the posterior induced by m (from log-concavity), $u \mapsto \beta(\mu^{-1}(u))$ is continuous and strictly increasing. Hence $H(\cdot, \pi)$ is continuous and strictly increasing in u, with

$$\frac{\partial H}{\partial u}(u, \pi) = (1-\alpha) - \beta_u(u) \;\geq\; 1 - \alpha - \sup_{\theta} B'(\theta) \;>\; 0,$$

and $\frac{\partial H}{\partial \pi}(u, \pi) = -2/\lambda < 0$. Because $\lim_{u \downarrow \underline{\theta}_0} H(u, \pi) = -\infty$ and $\lim_{u \uparrow \overline{\theta}_0} H(u, \pi) = +\infty$, the intermediate-value theorem gives a unique zero $u_{\mathrm{SM}}(\pi) \in (\underline{\theta}_0, \overline{\theta}_0)$. The implicit-function theorem then yields

$$\frac{d u_{\mathrm{SM}}}{d\pi} = -\frac{\partial H / \partial \pi}{\partial H / \partial u} = \frac{2/\lambda}{(1-\alpha) - \beta_u\big(u_{\mathrm{SM}}(\pi)\big)} \;>\; 0,$$

so $\pi \mapsto u_{\mathrm{SM}}(\pi)$ is strictly increasing.

(ii) Partial pooling for $\pi \leq \frac{1}{2}$. At $\pi = \frac{1}{2}$, (SM-cut) reduces to $(1-\alpha)u - \beta(\mu^{-1}(u)) = 0$, whose solution $u_{\mathrm{SM}}(\frac{1}{2}) \leq 0$ by the normalization in (P1)–(P3). Given $u \leq 0$, $\Lambda(\theta; u)$ has a unique zero $\theta^*(u)$ with $\underline{\theta}_0 < \theta^*(u) < u$: types below $\theta^*(u)$ find the minimally persuasive message too costly and are rejected; types in $[\theta^*(u), u)$ pool at $m^{\mathrm{req}}(u)$ and are approved; types above u tell the truth and are approved. For any lenient $\pi < \frac{1}{2}$, part (i) implies $u_{\mathrm{SM}}(\pi) < u_{\mathrm{SM}}(\frac{1}{2}) \leq 0$, and the same argument applies.

(iii) Critical quota and full separation. By Lemma 1 and the envelope properties of C, at the (unique) root $\theta^*(u)$ we have $\Lambda_\theta(\theta^*(u); u) > 0$ and $\Lambda_u(\theta^*(u); u) < 0$. Hence $\theta^*(u)$ is continuous and strictly increasing in u by the implicit-function theorem:

$$\frac{d\theta^*}{du} \;=\; -\frac{\Lambda_u(\theta^*(u); u)}{\Lambda_\theta(\theta^*(u); u)} \;>\; 0.$$

Let $L(u) := u - \theta^*(u)$. Part (ii) gives $L\big(u_{\mathrm{SM}}(\frac{1}{2})\big) > 0$. By Assumption **??**, for any fixed $\theta < u$, $C\big(m^{\mathrm{req}}(u), \theta\big) \to \infty$ as $u \uparrow \overline{\theta}_0$ (because $m^{\mathrm{req}}(u) - \theta \to \infty$ and the "talk-up" marginal cost is positive and increasing). Therefore $\Lambda(\theta; u) \to -\infty$ pointwise in $\theta < u$, which forces $\theta^*(u) \uparrow u$ (otherwise $\Lambda(\theta^*(u); u) < 0$ would contradict the definition of θ^*). Consequently, $L(u) \downarrow 0$ as $u \uparrow u_{\mathrm{SM}}(1)$. By continuity of $u_{\mathrm{SM}}(\cdot)$ and $L(\cdot)$, there exists at least one $\bar{\pi} \in (\frac{1}{2}, 1]$ with $L\big(u_{\mathrm{SM}}(\bar{\pi})\big) = 0$, i.e. $\theta^*\big(u_{\mathrm{SM}}(\bar{\pi})\big) = u_{\mathrm{SM}}(\bar{\pi})$, and the unique PSE is fully separating. If, in addition, $L'(u) < 0$ on $[u_{\mathrm{SM}}(\frac{1}{2}), u_{\mathrm{SM}}(1)]$ (e.g., under DWBR), then $L\big(u_{\mathrm{SM}}(\pi)\big)$ is strictly decreasing in π; hence $\bar{\pi}$ is unique.

Affine-B corollary. If $B(\theta) = b_0 + b_1\theta$ with $b_1 \in (0, 1-\alpha)$ then $\beta(\mu^{-1}(u)) = b_0 + b_1 u$, and solving (SM-cut) gives $u_{\mathrm{SM}}(\pi) = \big(-b_0 - (1-2\pi)/\lambda\big)/(1-\alpha-b_1)$ and $\frac{d u_{\mathrm{SM}}}{d\pi} = (2/\lambda)/(1-\alpha-b_1) > 0$. □

The intuition behind Proposition 6 is as follows: Once the requisite quota π is high enough, no type finds lying worthwhile. To see why, consider the net benefit to a shareholder from obtaining approval conditional on a voting rule. With very high quotas, controllers have to send a costly message that convinces almost all of the minority shareholders to approve the project, which dramatically lowers the marginal benefit from lying. Accordingly, the equilibrium becomes fully separating.

Proposition 7 (Strict but not-too-strict SM enlarges the pool)**.** *Let $\pi \in (0.5, \bar{\pi})$, where $\bar{\pi}$ is the critical quota defined in Proposition 6 (i). Denote the MoM belief cut-off by $u_{\text{MoM}} = 0$ and the SM cut-off by $u_{\text{SM}}(\pi) = -\dfrac{1-2\pi}{\lambda} > 0$.*

$$F\big(\theta^*(u_{\text{SM}}(\pi))\big) - F(\underline{\theta}_0) > F\big(\theta^*(u_{\text{MoM}})\big) - F(\underline{\theta}_0), \quad 0.5 < \pi < \bar{\pi}.$$

Hence the pooling probability (and interval) under a still-passable but strict super-majority quota exceeds that under MoM.

Proof. First, we establish the monotone shift of the lying-profit curve. For any u let

$$\Psi(u) := \underbrace{\Lambda(\underline{\theta}_0; u)}_{\text{bottom-support test}}$$

defined in Lemma 3. By Lemma 1, $\Psi'(u) = \partial\Lambda/\partial u < 0$ whenever the persuasive message $m^{\text{req}}(u) = \mu^{-1}(u)$ lies above $\underline{\theta}_0$. Since $u_{\text{SM}}(\pi) > 0$, $m^{\text{req}}(u_{\text{SM}}(\pi)) > \underline{\theta}_0$, so

$$\Psi\big(u_{\text{SM}}(\pi)\big) = \Lambda\big(\underline{\theta}_0; u_{\text{SM}}(\pi)\big) < \Lambda(\underline{\theta}_0; 0) = \Psi(u_{\text{MoM}}). \tag{1}$$

By Lemma 2, $d\theta^*/du > 0$ whenever $m^{\text{req}}(u) > \underline{\theta}_0$. Since $u_{\text{SM}}(\pi) > u_{\text{MoM}}$,

$$\theta^*\big(u_{\text{SM}}(\pi)\big) > \theta^*\big(u_{\text{MoM}}\big). \tag{2}$$

Because F is strictly increasing on the support, applying F to (2) gives

$$F\big(\theta^*(u_{\text{SM}}(\pi))\big) - F(\underline{\theta}_0) > F\big(\theta^*(u_{\text{MoM}})\big) - F(\underline{\theta}_0),$$

as required. Finally, $\pi < \bar{\pi}$ implies $\Psi\big(u_{\text{SM}}(\pi)\big) < 0$ by Proposition 6(i). Lemma 3 then asserts that a negative Ψ places the cut-off in the *partial-pooling* regime, whereas $\Psi \geq 0$ would yield full separation. Because $u_{\text{MoM}} = 0$ also satisfies $\Psi(0) < 0$, both rules we compare indeed support partial pooling, so the ordering of pooling probabilities established above is valid. □

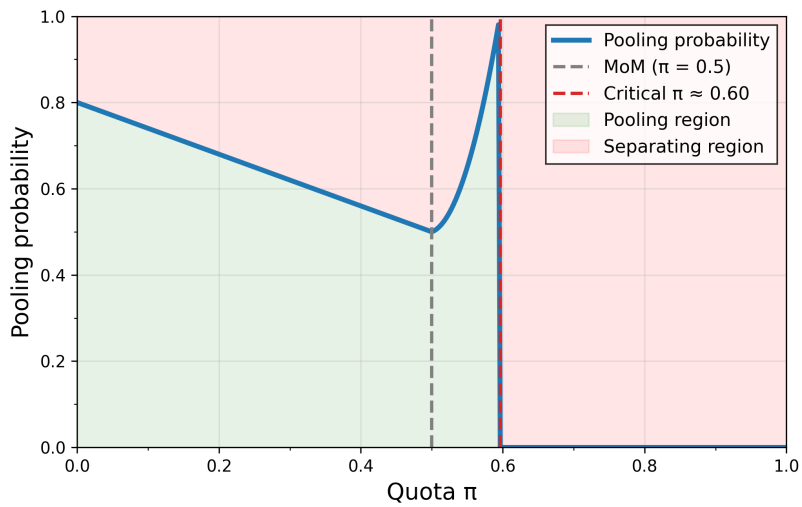
The intuition for Proposition 7 is somewhat counterintuitive and lies with the properties of the asymmetric convex cost function in the model. Recall that $C(m, \theta)$ is convex and has a positive marginal cost only for upward deviations. For many intermediate types, the costs of messaging explode in magnitude relative to the linear increase in benefits from having their projects approved. For instance, an intermediate-type CS may experience a cost increase of about $50 - 100\%$ in equilibrium under a stringent (but not too stringent) SM regime, while facing only a $20 - 30\%$ increase in payoffs, conditional on having the project approved. This disproportionate rise in the costs required penalize mid-to-low types disproportionately, increasing the band of controllers who pool.

Figure 1 illustrates the lying–profit curves and the associated pooling probabilities for each voting

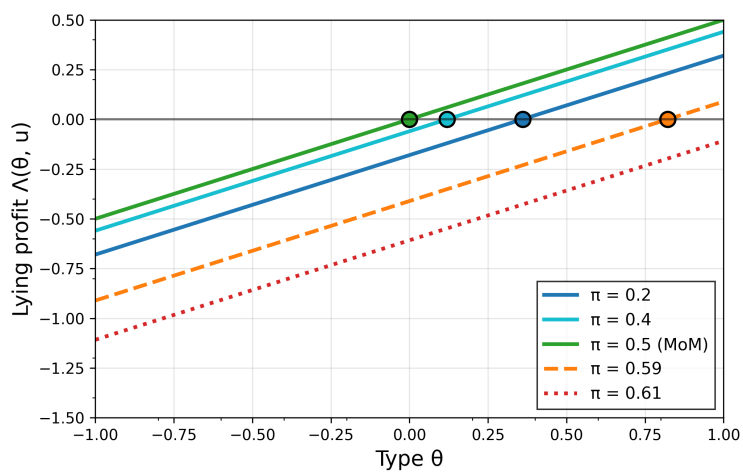
regime. Under the parameters chosen in the simulation, pooling probabilities decline from the lower end of the support and reach their minimum at the MoM threshold $\pi = 0.5$. In Panel A they rise steadily until the critical quota $\pi = \bar{\pi}$, after which they collapse to zero, in line with Propositions **??**, 6, and 7. Panel B displays the lying–profit curves for several values of π. Once π exceeds $\bar{\pi}$, the lying–profit function no longer satisfies the single-crossing property on the entire support; by the bottom-support test in Lemma 3, this implies complete separation.

Figure 1: Pooling Probabilities and Lying-Profit Curves

(a) Panel A: Pooling Probabilities



(b) Panel B: Lying-Profit Curves



Panel A displays the simulated pooling probability (solid blue) for a controller with an objective function $\mathrm{CS}(\theta, m) = (\alpha+\beta)\theta - C(m)$, $C(m) = \kappa\big(\max\{m, 0\}\big)^2 + \eta|m| = \eta(-m)$ if $m \leq 0$, $\kappa m^2 + \eta m$ if $m > 0$, as a function of the minority quota π for the parameter set $(\alpha, \beta, \lambda) = (0.4, 0.10, 1)$. The grey dashed line marks the Majority-of-the-Minority threshold $\pi = 0.5$, while the red dashed line indicates the critical quota $\pi \approx 0.60$ at which pooling vanishes. Panel B plots the lying-profit function $\Lambda(\theta, u)$ for five representative quotas $\pi \in \{0.2, 0.4, 0.5, 0.59, 0.61\}$; solid curves correspond to quotas that still permit pooling, dashed/dotted curves to quotas with complete separation, and the open markers highlight the type cut-off θ^* where Λ crosses zero.

5 Ex-ante welfare: MoM versus lenient and strict SM

To evaluate total shareholder welfare under a given rule, I measure *ex ante total surplus* before the controller's type is realized. Expectations are taken over the prior $F(\theta)$ and—under super-majority rules—over the idiosyncratic minority biases b_i. The first best ("FB") approves a project iff $\theta \geq 0$ and incurs no signaling cost.

Given u, types are divided as follows:

$$\begin{array}{llll} \text{Rejected (truthful):} & \theta \in [\underline{\theta}_0, \theta^*(u)) & \Rightarrow & m = \theta, \quad C(\theta, \theta) = 0; \\ \text{Approved (persuaders):} & \theta \in [\theta^*(u), u) & \Rightarrow & m = m_{\text{req}}(u), \quad C\big(m_{\text{req}}(u), \theta\big) > 0; \\ \text{Approved (truthful):} & \theta \in [u, \overline{\theta}_0] & \Rightarrow & m = \theta, \quad C(\theta, \theta) = 0. \end{array}$$

Thus all signaling costs arise *only* on the persuader region $[\theta^*(u), u)$.

5.1 Welfare Decomposition

Let $\phi(\theta) := \theta + B(\theta)$ denote total surplus if implemented at type θ (If transfers are zero-sum, then $\phi(\theta) := \theta$). Define:

$$\text{(signaling)} \qquad S(u) := \int_{\theta^*(u)}^{u} C\big(m_{\text{req}}(u), \theta\big)\, f(\theta)\, d\theta,$$

$$\text{(Type II: negative projects approved)} \qquad \tau_1(u) := \begin{cases} -\displaystyle\int_{\theta^*(u)}^{0} \phi(\theta)\, f(\theta)\, d\theta, & \theta^*(u) < 0, \\ 0, & \theta^*(u) \geq 0, \end{cases}$$

$$\text{(Type I: positive projects rejected)} \qquad \tau_2(u) := \begin{cases} 0, & \theta^*(u) \leq 0, \\ \displaystyle\int_{0}^{\theta^*(u)} \phi(\theta)\, f(\theta)\, d\theta, & \theta^*(u) > 0. \end{cases}$$

Convention: integrals over empty intervals are zero. When $u \leq 0$ (lenient SM or MoM) we have $\theta^*(u) < 0$, so $\tau_2(u) = 0$ and only Type II errors (and signaling) can occur. Under sufficiently strict SM, $\theta^*(u)$ may become positive, in which case $\tau_2(u) > 0$.

Let

$$W_{\text{FB}} := \int_{0}^{\overline{\theta}_0} \phi(\theta)\, f(\theta)\, d\theta$$

be the first-best benchmark. Ex ante welfare under cut-off u is thus

$$W(u) = W_{\text{FB}} - S(u) - \tau_1(u) - \tau_2(u). \tag{W1}$$

5.2 Welfare Comparisons

Proposition 8 (Ex-ante welfare: MoM versus lenient and strict SM)**.** *Maintain Assumptions in Section 2 with* $\underline{\theta}_0 < 0 < \overline{\theta}_0$ *and* $B(\underline{\theta}_0) = 0$*. Let*

$$u_{len}(\pi) = -\frac{1-2\pi}{\lambda}\ (<0), \qquad u_{MoM} = 0, \qquad u_{str}(\pi) = \frac{2\pi-1}{\lambda}\ (>0)$$

be the interior belief cut-offs for a lenient quota $\pi < 0.5$*, MoM, and a* passable *strict quota* $\pi \in (0.5, \bar{\pi})$*, respectively. Denote by*

$$W_{pool} := W\big(u_{str}(\pi)\big), \qquad W_{len} := W\big(u_{len}(\pi)\big), \qquad W_{MoM} := W(0), \qquad W_{FB} := \int_0^{\overline{\theta}_0} \phi(\theta)\, f(\theta)\, d\theta$$

the ex-ante welfare under (i) the pooling equilibrium of the strict rule, (ii) the lenient rule, (iii) MoM, and (iv) the first best, computed from (W1)*.*

A strict rule may in addition support a fully separating equilibrium; let $q \in [0,1]$ *be the probability (determined by an equilibrium selection device) that separation rather than pooling occurs. Expected welfare under the strict rule is then*

$$W_{str}(q) \;=\; q\, W_{FB} \;+\; (1-q)\, W_{pool}. \tag{W-S}$$

(i) ***MoM beats each single-equilibrium SM.*** $W_{MoM} > W_{len}$ *and* $W_{MoM} > W_{pool}$*.*

(ii) ***Lenient versus pooling–strict SM.*** *The ordering depends on the relative size of signalling and error losses:*

$$W_{len} \begin{cases} > W_{pool}, & \text{if } S_{len} - S_{pool} \;<\; \tau_{2,pool} - \tau_{1,len}, \\ = W_{pool}, & \text{if the two sides coincide,} \\ < W_{pool}, & \text{otherwise.} \end{cases}$$

(iii) ***Strict SM with selection probability*** q***.*** *Define*

$$\bar{q} := \frac{W_{MoM} - W_{pool}}{W_{FB} - W_{pool}} \in (0,1).$$

Then

$$W_{str}(q) \begin{cases} < W_{MoM}, & q < \bar{q}, \\ = W_{MoM}, & q = \bar{q}, \\ > W_{MoM}, & q > \bar{q}. \end{cases}$$

Thus MoM maximises welfare whenever the fully separating branch of the strict rule is selected with probability below $\bar{q}$*.*

Proof. First, we examine the signalling loss $S(u)$. For every interior rule the equilibrium satisfies $\theta^*(u) < u$, so the pooling set is non-empty and incurs positive signalling cost $S(u) > 0$. Propositions 4 (lenient vs MoM) and 7 (strict vs MoM) give the strict ordering

$$S_{\text{len}} > S_{\text{MoM}} < S_{\text{pool}} \tag{S-order}$$

Second, we examine the mis-classification losses, which induce Type I and Type II errors.

- *Lenient SM, $u = u_{len} < 0$.* Only type-II error applies; therefore $\tau_{1,\text{len}} > 0 = \tau_{1,\text{MoM}} = \tau_{1,\text{pool}}$ and $\tau_{2,\text{len}} = \tau_{2,\text{MoM}} = \tau_{2,\text{pool}} = 0$.
- *Pooling strict SM, $u = u_{pool} > 0$.* Only type-I error applies; therefore $\tau_{2,\text{pool}} > 0 = \tau_{2,\text{MoM}} = \tau_{2,\text{len}}$ and $\tau_{1,\text{pool}} = \tau_{1,\text{MoM}} = \tau_{1,\text{len}} = 0$.

In summary,

$$\tau_{1,\text{len}} > 0 = \tau_{1,\text{MoM}} = \tau_{1,\text{pool}}, \qquad \tau_{2,\text{pool}} > 0 = \tau_{2,\text{MoM}} = \tau_{2,\text{len}} \tag{T-order}$$

Comparing the welfare of the two pooling rules, we have $W(u) = W_{\text{FB}} - S(u) - \tau_1(u) - \tau_2(u)$. Inserting (S-order)–(T-order) yields

$$W_{\text{MoM}} > W_{\text{len}} \quad \text{and} \quad W_{\text{MoM}} > W_{\text{pool}}, \tag{rank-pool}$$

since MoM has the lowest signalling cost and incurs *no* classification error. The difference between the two pooling rules is

$$W_{\text{len}} - W_{\text{pool}} = \left[S_{\text{pool}} - S_{\text{len}}\right] + \left[\tau_{2,\text{pool}} - \tau_{1,\text{len}}\right].$$

Hence

$$W_{\text{len}} > W_{\text{pool}} \iff S_{\text{len}} - S_{\text{pool}} < \tau_{2,\text{pool}} - \tau_{1,\text{len}},$$

exactly the condition stated in part (ii) of the proposition.

Finally, we examine the welfare of Strict SM with a selection probability q. Recall $W_{\text{str}}(q) = q\, W_{\text{FB}} + (1-q)\, W_{\text{pool}}$. Because $W_{\text{FB}} > W_{\text{MoM}} > W_{\text{pool}}$ by (rank-pool), there exists a unique

$$\bar{q} := \frac{W_{\text{MoM}} - W_{\text{pool}}}{W_{\text{FB}} - W_{\text{pool}}} \in (0,1)$$

such that $W_{\text{str}}(\bar{q}) = W_{\text{MoM}}$. Linearity in q immediately gives

$$W_{\text{str}}(q) \begin{cases} < W_{\text{MoM}}, & q < \bar{q}, \\ = W_{\text{MoM}}, & q = \bar{q}, \\ > W_{\text{MoM}}, & q > \bar{q}, \end{cases}$$

completing the proof. □

Intuitively, the first-best benchmark approves every project whose fundamental value is non–negative and incurs no signaling outlay. By contrast, any super-majority rule shifts the belief cut-off u away from zero, enlarging the set of types that must exaggerate and thereby raising the real resources burned on persuasion. A lenient quota pushes the cut-off into negative territory; this induces additional low-quality projects to pass (type-II error) and enlarges the pooling set, so both the misallocation loss and the signalling loss rise relative to MoM. A *passable* strict quota moves the cut-off above zero; now some positive-NPV projects are turned away (type-I error) while the incentive to exaggerate becomes even stronger, again lowering welfare below MoM as long as the game remains on its pooling equilibrium. If the strict rule happens to admit a fully separating equilibrium, expected welfare lies on a linear segment between the pooling outcome and the first best; only when the probability of landing on this first-best branch exceeds the explicit threshold $\bar{q}$ can the strict rule overtake MoM. Thus, the MoM strictly dominates both single-equilibrium super-majorities and remains welfare-superior to a mixed strict rule unless the market coordinates on full separation with sufficiently high probability.

6 Conclusion

This paper has shown that the majority of the minority (MoM) rule exhibits superior informational properties compared to other voting thresholds in conflicted transactions. By developing a model where misrepresentations that harm minorities carry litigation risk while conservative reporting is relatively cheap, I demonstrate that MoM occupies a ”knife-edge” position: it minimizes the pooling equilibrium region where truthful and deceptive communications cannot be distinguished. While sufficiently stringent supermajorities can achieve complete separation, they block valuable transactions. These findings provide a novel informational rationale for MoM’s widespread use and suggest that Delaware’s SB21 reforms, which weakened MoM requirements, may reduce disclosure quality in controller transactions.

Several extensions merit future research. First, endogenous ownership structures would illuminate how anticipation of voting rules affects initial control concentration—controllers might adjust stakes to exploit lenient thresholds or avoid stringent ones. Second, strategic voting by minorities could reveal how sophisticated shareholders coordinate, acquire information, or condition votes on market signals, potentially reinforcing or attenuating different rules’ informational properties. Third, allowing endogenous investment levels or multidimensional project characteristics could show how voting rules affect not just whether transactions occur, but their design—controllers might distort projects to secure approval. Fourth, incorporating intermediaries like proxy advisors whose reputational incentives shape information transmission could provide a richer understanding of disclosure dynamics. Finally, the model’s empirical predictions—that MoM produces more informative disclosures than very lenient or stringent alternatives—await testing through cross-jurisdictional or temporal variation in voting rules.

As controlling shareholders dominate global corporate ownership, understanding how voting rules shape their communication with minorities becomes increasingly important. My work demonstrates that voting rules do more than aggregate preferences—they fundamentally deter-

mine what information gets produced and transmitted in conflicted transactions.

A Appendix

A.1 Additional Proofs

Corollary 1 (Degenerate voting rules)**.** *Let $u = \underline{\theta}$ be the rule–specific belief cut-off.*

(i) ***Trivially easy rule*** $\left[u < \underline{\theta}_0\right]$*. Every posterior belief satisfies $\mu(m) \geq u$, hence minorities always vote "Yes". For the controller any message in the approval region is accepted; because $C(m, \theta) \geq 0$ with equality at $m = \theta$, truthful reporting maximises payoff but* any *message weakly below the truth is also a best reply. Consequently an uncountable continuum of Perfect–Sequential Equilibria exists, ranging from complete pooling $\sigma(\theta) \equiv m_L$ (with any $m_L \leq u$) to full separation $\sigma(\theta) = \theta$. The pooling probability can take any value in* $[0, 1]$.

(ii) ***Trivially tough rule*** $\left[u > \overline{\theta}_0\right]$*. No posterior belief can reach the hurdle, so minorities always vote "No". The controller's unique best reply is to choose the cheapest message, i.e. $\sigma(\theta) = \theta$ (because $C(m, \theta)$ is minimised at $m = \theta$). The unique PSE is therefore fully separating with universal rejection.*

Lemma 5 (No single-message pooling)**.** *Fix a belief cut-off $\underline{\theta}$ that lies strictly inside the support, i.e. $\underline{\theta}_0 < \underline{\theta} < \overline{\theta}_0$. There is* no *Perfect–Sequential Equilibrium in which every type $\theta \in (\underline{\theta}_0, \overline{\theta}_0)$ sends the* same *message $m^\flat$, regardless of whether minorities approve or reject that message.*

Proof. Denote by $m^{\text{req}} := \mu^{-1}(\underline{\theta})$ the smallest message that induces minority approval. Two cases:

(a) Common message induces rejection: $\mu(m^\flat) < \underline{\theta}$. Then every minority votes "No" and each controller earns $U^{\text{eq}}(\theta) = -C(m^\flat, \theta)$.

Pick a high type $\bar{\theta}$ close enough to $\overline{\theta}_0$ that

$$\alpha\bar{\theta} + B(\bar{\theta}) \;>\; C\left(m^{\text{req}}, \bar{\theta}\right). \tag{$*$}$$

(Feasible because B is strictly increasing and $C(m, \theta)$ weakly *decreases* in θ by Assumption S4.)

If $\bar{\theta}$ deviates to $m^{\text{req}} \geq \underline{\theta}$ the project is approved and her payoff becomes

$$U^{\text{dev}} = \alpha\bar{\theta} + B(\bar{\theta}) - C\left(m^{\text{req}}, \bar{\theta}\right) > 0 \;\text{ by } (*).$$

Because $C(\theta, \theta) = 0$, the equilibrium payoff satisfies $U^{\text{eq}}(\bar{\theta}) = -C(m^\flat, \bar{\theta}) \leq 0$. Hence $U^{\text{dev}} > U^{\text{eq}}$, contradicting sequential rationality.

(b) Common message induces approval: $\mu(m^\flat) \geq \underline{\theta}$. Then every minority votes "Yes" and the equilibrium payoff is

$$U^{\text{eq}}(\theta) = \alpha\theta + B(\theta) - C(m^\flat, \theta).$$

Because $C_m(m, \theta) < 0$ when $m < \theta$ (Assumption S3), any type can reduce her signalling cost by under-reporting slightly while remaining above the cut-off. Choose $\theta^\dagger$ such that $m^\flat > \theta^\dagger > \underline{\theta}$ and set $m^\dagger := \theta^\dagger$. Then

$$C(m^\dagger, \theta^\dagger) < C(m^\flat, \theta^\dagger),$$

and minorities still approve because $\mu(m^\dagger) = \theta^\dagger \geq \underline{\theta}$. The deviating payoff $U^{\text{dev}} = \alpha\theta^\dagger + B(\theta^\dagger) - C(m^\dagger, \theta^\dagger)$ is strictly higher than $U^{\text{eq}}(\theta^\dagger)$, so the pooling profile fails to be an equilibrium.

Since both rejection-pooling and approval-pooling are ruled out, no single-message pooling equilibrium exists when the cut-off lies strictly inside the support. □

A.2 Credible-Beliefs Refinement (Grossman–Perry 1986)

A.2.1 Definition

Consider any signalling game with Perfect Bayesian Equilibria (PBE). Let m_{off} be a *message that is not sent with positive probability in the candidate equilibrium.* A PBE satisfies the credible-beliefs refinement if the receiver's belief $\mu(\cdot \mid m_{\text{off}})$ assigns probability 1 to the set

$$T^{\text{best}}(m_{\text{off}}) := \Big\{ \theta \in \Theta : \; U_S\big(m_{\text{off}}, \theta\big) \;\geq\; U_S\big(m', \theta\big) \;\; \forall m' \text{ in the equilibrium support} \Big\},$$

i.e. the set of sender types that would (weakly) benefit *most* from deviating to m_{off}. If T^{best} is empty, beliefs are unrestricted.

Intuitively, the receiver should deem the unexpected signal most likely to come from the type with the strongest incentive to deviate, which is usually the *worst* type from the receiver's standpoint. This pessimism discourages profitable deviations and helps select a unique equilibrium.

A.2.2 Illustration in Our Partial-Pooling Equilibrium

Equilibrium recap. In the pooling band where $\theta < \theta^*$, all types send the common message m_L and minority shareholders reject the proposal. In the separating band where $\theta \geq \theta^*$, each type reports truthfully; the minorities approve precisely when $(1-\alpha)\theta - B(\theta) \geq \underline{\kappa}^{(\lambda)}$.

Off-path deviations. Consider a sender who instead transmits a higher message m_H that is not prescribed by the equilibrium. High-quality types ($\theta \geq \theta^*$) already secure approval without cost, so switching to m_H cannot improve their payoff. Low-quality types ($\theta < \theta^*$) are currently rejected and would gain $\alpha\theta + B(\theta) - C(m_H, \theta)$ only if the deviation were accepted. Hence the set of types with the strongest incentive to deviate, $T^{\text{best}}(m_H)$, contains exactly these low types.

Credible beliefs assign probability one to $\theta \in T^{\text{best}}(m_H)$ upon observing m_H. Minority shareholders therefore adopt the most pessimistic posterior, and the inequality $(1-\alpha)\mu(m_H) - \beta(m_H) \geq \underline{\kappa}^{(\lambda)}$

fails. They vote "No." The deviating sender then pays the full signalling cost $C(m_H, \theta)$ without gaining private benefits, making deviation strictly unprofitable.

Because every off-equilibrium message is treated in this way, no type wishes to deviate, and the partial-pooling profile remains the unique perfect sequential equilibrium under the given voting rule. Credible beliefs thus eliminate the need for arbitrary off-path assumptions: outsiders always interpret surprise signals in the manner most damaging to the sender, thereby ruling out equilibria that depend on optimistic beliefs and resolving multiplicity.

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